Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: BWC Weststeyn Dairy LP

Physical address of dairy:

1763 S Hewitt RD	Linden	San Joaquin	95236
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1977

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X183-X300-XX05-XXXX X183-X300-XX06-XXXX X183-X300-XX07-XXXX X183-X300-XX08-XXXX

B. OPERATORS

Weststeyn, Bert

Operator name: Weststeyn, Bert Telephone no.: (209) 404-9100 (Landline) (Cellular)

1763 S Hewitt RD	Linden	CA	95236
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Weststeyn, Bert

Legal owner name: Weststeyn, Bert Telephone no.: (209) 404-9100 (Landline) (Cellular)

1763 S Hewitt RD	Linden	CA	95236
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	200	1,220	1,000	450	580
Number under roof	2,800	200	0	0	0	0
Maximum number	2,800	400	1,220	1,000	450	580
Average number	2,800	400	1,220	1,000	450	580
Avg live weight (lbs)	1,400	1,450	950	700		

Predominant milk cow breed: Holstein

Average milk production: 80 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 106,567.12 *tons per reporting period*

Total nitrogen from manure: 1,318,340.16 *lbs per reporting period*

After ammonia losses (30% loss applied): 922,838.11 *lbs per reporting period*

Total phosphorus from manure: 215,889.10 *lbs per reporting period*

Total potassium from manure: 554,621.83 *lbs per reporting period*

Total salt from manure: 1,410,360.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 89,444,240 *gallons*

Total nitrogen generated: 325,826.78 *lbs*

Total phosphorus generated: 43,949.09 *lbs*

Total potassium generated: 372,633.18 *lbs*

Total salt generated: 1,811,854.56 *lbs*

	89,444,240	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	89,444,240	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Deep Well #10	Ground water
Deep Well #15	Ground water
District H20	Ground water
IW 10	Ground water
IW 13	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
04/28/2023	Solid commercial fertilizer *UN 32*	27.72 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000
07/02/2023	Solid commercial fertilizer *UN 32*	46.97 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	47,753.80	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	47,753.80	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/10/2023	Compost	12,501.00 *ton*	Dry-weight	32.9		23,900.00	10,700.00	14,700.00		55.90

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	400,954.57	179,506.86	246,612.23	9,377,975.18
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	400,954.57	179,506.86	246,612.23	9,377,975.18

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Dondero 25-14	63	63	2	both	X183-X290-XX06-XXXX
Dondero 25-15	63	63	2	both	X183-X290-XX06-XXXX
Dondero 25-19	50	50	2	manure	X183-X290-XX06-XXXX
Dondero 25-4	60	60	2	manure	X183-X220-XX01-XXXX
Field 1	67	67	2	process wastewater	X183-X300-XX08-XXXX
Field 1S	76	76	2	manure	X187-X030-XX02-XXXX
Field 2	36	36	2	process wastewater	X183-X300-XX08-XXXX
Field 3	75	75	2	process wastewater	X183-X300-XX06-XXXX X183-X310-XX06-XXXX
Field 3S	53	53	2	manure	X187-X030-XX02-XXXX
Field 4	88	88	2	process wastewater	X183-X310-XX06-XXXX
Field 5	50	50	2	process wastewater	X183-X310-XX04-XXXX
Field 6	50	50	2	process wastewater	X183-X310-XX04-XXXX
Field 7	73	73	2	process wastewater	X183-X300-XX05-XXXX X183-X310-XX04-XXXX
Field 8	79	79	2	manure	X183-X370-XX11-XXXX X183-X370-XX28-XXXX
Totals for areas that were used for application	883	883	28		
Totals for areas that were not used for application					
Land application area totals	883	883	28		

B. CROPS AND HARVESTS

Dondero 25-14

Field name: Dondero 25-14

Dondero 25-14

10/08/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 63 Plant date: 10/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/14/2023	2,740.00 *ton*	As-is		77.8	5,860.00	670.00	7,400.00		14.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	425.00	62.50	410.00	0.00
Total actual harvest content	43.49	509.73	58.28	643.68	2,800.02

06/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 63 Plant date: 06/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/14/2023	1,900.00 *ton*	As-is		71.8	3,270.00	570.00	5,040.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.16	197.24	34.38	304.00	1,445.81

Dondero 25-15

Field name: Dondero 25-15

10/08/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 63 Plant date: 10/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/12/2023	2,748.00 *ton*	As-is		73.0	6,990.00	980.00	9,820.00		18.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	425.00	62.50	410.00	0.00
Total actual harvest content	43.62	609.79	85.49	856.68	4,451.76

Dondero 25-15

07/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 63 Plant date: 07/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	2,025.00 *ton*	As-is		70.6	3,760.00	690.00	3,890.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	32.14	241.71	44.36	250.07	1,171.80

Dondero 25-19

Field name: Dondero 25-19

10/22/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 50 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	1,220.00 *ton*	As-is		70.4	3,760.00	600.00	6,120.00		11.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	425.00	62.50	410.00	0.00
Total actual harvest content	24.40	183.49	29.28	298.66	1,603.37

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	1,887.50 *ton*	As-is		67.8	3,770.00	2,700.00	4,520.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	37.75	284.64	203.85	341.26	1,142.62

Dondero 25-4

Field name: Dondero 25-4

10/20/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 60 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	1,413.00 *ton*	As-is		72.4	4,250.00	700.00	6,450.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	425.00	62.50	410.00	0.00
Total actual harvest content	23.55	200.18	32.97	303.80	1,468.95

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 60 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	2,087.50 *ton*	As-is		66.2	3,950.00	1,360.00	5,440.00		2.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	34.79	274.85	94.63	378.53	635.02

Field 1

Field name: Field 1

10/08/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 67 Plant date: 10/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	1,828.00 *ton*	As-is		67.0	4,220.00	720.00	7,340.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	425.00	62.50	410.00	0.00
Total actual harvest content	27.28	230.27	39.29	400.52	1,908.76

Field 1

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 67 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	2,118.80 *ton*	As-is		65.4	3,630.00	680.00	6,280.00		3.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	31.62	229.59	43.01	397.20	656.51

Field 1S

Field name: Field 1S

10/17/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 76 Plant date: 10/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/30/2023	1,460.00 *ton*	As-is		56.2	5,050.00	1,100.00	10,200.00		8.97

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	425.00	62.50	410.00	0.00
Total actual harvest content	19.21	194.03	42.26	391.89	1,509.51

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 76 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	2,837.50 *ton*	As-is		65.8	3,630.00	1,480.00	7,010.00		2.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	37.34	271.06	110.51	523.44	663.98

Field 2

Field name: Field 2

10/10/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 36 Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	997.00 *ton*	As-is		71.6	6,990.00	1,050.00	8,740.00		12.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	425.00	62.50	410.00	0.00
Total actual harvest content	27.69	387.17	58.16	484.10	1,982.04

05/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 05/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	1,287.50 *ton*	As-is		65.2	3,970.00	680.00	6,480.00		2.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	35.76	283.97	48.64	463.50	522.73

Field 3

Field name: Field 3

10/08/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 75 Plant date: 10/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/08/2023	3,146.00 *ton*	As-is		68.2	8,110.00	1,360.00	18,400.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	425.00	62.50	410.00	0.00
Total actual harvest content	41.95	680.37	114.09	1,543.64	2,507.74

Field 3

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	2,437.50 *ton*	As-is		63.6	4,620.00	620.00	4,620.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	32.50	300.30	40.30	300.30	1,277.64

Field 3S

Field name: Field 3S

10/18/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 53 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	1,135.00 *ton*	As-is		64.0	5,540.00	1,180.00	9,700.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	425.00	62.50	410.00	0.00
Total actual harvest content	21.42	237.28	50.54	415.45	1,572.72

06/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 53 Plant date: 06/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/23/2023	1,925.00 *ton*	As-is		66.8	3,620.00	610.00	4,530.00		3.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	36.32	262.96	44.31	329.07	795.86

Field 4

Field name: Field 4

10/12/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 88 Plant date: 10/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/15/2023	3,822.00 *ton*	As-is		79.3	5,650.00	910.00	12,400.00		13.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	425.00	62.50	410.00	0.00
Total actual harvest content	43.43	490.78	79.05	1,077.11	2,499.33

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 88 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	2,575.00 *ton*	As-is		67.8	4,670.00	640.00	4,510.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	29.26	273.30	37.45	263.94	979.90

Field 5

Field name: Field 5

10/15/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 50 Plant date: 10/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/06/2023	2,054.00 *ton*	As-is		68.2	5,990.00	1,170.00	10,700.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	425.00	62.50	410.00	0.00
Total actual harvest content	41.08	492.14	96.13	879.11	2,926.21

Field 5

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	1,362.50 *ton*	As-is		59.8	5,070.00	1,200.00	6,780.00		1.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.25	276.32	65.40	369.51	284.82

Field 6

Field name: Field 6

10/08/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 50 Plant date: 10/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/08/2023	2,282.00 *ton*	As-is		73.8	5,080.00	1,070.00	12,200.00		14.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	425.00	62.50	410.00	0.00
Total actual harvest content	45.64	463.70	97.67	1,113.62	3,348.15

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	1,762.50 *ton*	As-is		70.8	3,420.00	600.00	4,760.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	35.25	241.11	42.30	335.58	1,420.43

Field 7

Field name: Field 7

10/08/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 73 Plant date: 10/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
06/08/2023	3,176.00 *ton*	As-is		73.4	5,020.00	930.00	8,670.00		12.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	425.00	62.50	410.00	0.00
Total actual harvest content	43.51	436.81	80.92	754.41	2,962.64

06/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 73 Plant date: 06/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	2,425.00 *ton*	As-is		67.4	4,210.00	550.00	5,390.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	33.22	279.71	36.54	358.10	1,407.83

Field 8

Field name: Field 8

10/16/2022: Rye Grass Silage

Crop: Rye Grass Silage Acres planted: 60 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	1,592.00 *ton*	As-is		67.2	5,150.00	820.00	7,380.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	425.00	62.50	410.00	0.00
Total actual harvest content	26.53	273.29	43.51	391.63	2,158.33

Field 8

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	1,587.50 *ton*	As-is		66.8	3,850.00	1,270.00	5,310.00		2.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	31.75	244.48	80.65	337.19	590.30

NUTRIENT BUDGET

A. LAND APPLICATIONS

Dondero 25-14 - 10/08/2022: Rye Grass Silage

Field name: Dondero 25-14

Crop: Rye Grass Silage

Plant date: 10/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	268.87	26.03	328.66	1,373.41	2,579,220.00 *gal*
Application event totals		268.87	26.03	328.66	1,373.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	5.67	0.00	0.00	915.57	7,200,000.00 *gal*
Application event totals		5.67	0.00	0.00	915.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	38.40	0.00	0.00	0.00	
Application event totals		38.40	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	5.50	0.00	0.00	886.95	6,975,000.00 *gal*
Application event totals		5.50	0.00	0.00	886.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	163.53	109.02	97.17	4,692.60	630.00 *ton*
Application event totals		163.53	109.02	97.17	4,692.60	

Dondero 25-14 - 06/27/2023: Corn, silage

Field name: Dondero 25-14

Crop: Corn, silage

Plant date: 06/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	214.49	94.42	392.85	7,134.17	600.00 *ton*
Application event totals		214.49	94.42	392.85	7,134.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.98	0.00	0.00	1,125.38	8,850,000.00 *gal*
Application event totals		6.98	0.00	0.00	1,125.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.68	0.00	0.00	1,077.70	8,475,000.00 *gal*
Application event totals		6.68	0.00	0.00	1,077.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.62	0.00	0.00	1,068.16	8,400,000.00 *gal*
Application event totals		6.62	0.00	0.00	1,068.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.56	0.00	0.00	1,058.62	8,325,000.00 *gal*
Application event totals		6.56	0.00	0.00	1,058.62	

Dondero 25-14 - 06/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.50	0.00	0.00	1,049.09	8,250,000.00 *gal*
Application event totals		6.50	0.00	0.00	1,049.09	

09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.38	0.00	0.00	1,030.01	8,100,000.00 *gal*
Application event totals		6.38	0.00	0.00	1,030.01	

Dondero 25-15 - 10/08/2022: Rye Grass Silage

Field name: Dondero 25-15

Crop: Rye Grass Silage

Plant date: 10/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	278.66	26.98	340.62	1,423.39	2,673,090.00 *gal*
Application event totals		278.66	26.98	340.62	1,423.39	

05/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	5.79	0.00	0.00	934.64	7,350,000.00 *gal*
Application event totals		5.79	0.00	0.00	934.64	

05/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	38.40	0.00	0.00	0.00	
Application event totals		38.40	0.00	0.00	0.00	

Dondero 25-15 - 10/08/2022: Rye Grass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	5.62	0.00	0.00	906.03	7,125,000.00 *gal*
Application event totals		5.62	0.00	0.00	906.03	

10/23/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	163.53	109.02	97.17	4,692.60	630.00 *ton*
Application event totals		163.53	109.02	97.17	4,692.60	

Dondero 25-15 - 07/03/2023: Corn, silage

Field name: Dondero 25-15

Crop: Corn, silage

Plant date: 07/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	214.49	94.42	392.85	7,134.17	600.00 *ton*
Application event totals		214.49	94.42	392.85	7,134.17	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.98	0.00	0.00	1,125.38	8,850,000.00 *gal*
Application event totals		6.98	0.00	0.00	1,125.38	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.74	0.00	0.00	1,087.23	8,550,000.00 *gal*
Application event totals		6.74	0.00	0.00	1,087.23	

Dondero 25-15 - 07/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.62	0.00	0.00	1,068.16	8,400,000.00 *gal*
Application event totals		6.62	0.00	0.00	1,068.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.56	0.00	0.00	1,058.62	8,325,000.00 *gal*
Application event totals		6.56	0.00	0.00	1,058.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.50	0.00	0.00	1,049.09	8,250,000.00 *gal*
Application event totals		6.50	0.00	0.00	1,049.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.38	0.00	0.00	1,030.01	8,100,000.00 *gal*
Application event totals		6.38	0.00	0.00	1,030.01	

Dondero 25-19 - 10/22/2022: Rye Grass Silage

Field name: Dondero 25-19

Crop: Rye Grass Silage

Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Dondero 25-19 - 10/22/2022: Rye Grass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/26/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost	Compost	242.11	93.10	526.66	6,414.72	500.00 *ton*
Application event totals		242.11	93.10	526.66	6,414.72	

Dondero 25-19 - 06/09/2023: Corn, silage

Field name: Dondero 25-19

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	261.25	115.01	478.49	8,689.42	580.00 *ton*
Application event totals		261.25	115.01	478.49	8,689.42	

06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Deep Well #15	Ground water	8.95	0.00	0.00	1,318.54	6,930,000.00 *gal*
Application event totals		8.95	0.00	0.00	1,318.54	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Deep Well #15	Ground water	8.68	0.00	0.00	1,278.59	6,720,000.00 *gal*
Application event totals		8.68	0.00	0.00	1,278.59	

07/13/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Dondero 25-19 - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Deep Well #15	Ground water	8.68	0.00	0.00	1,278.59	6,720,000.00 *gal*
Application event totals		8.68	0.00	0.00	1,278.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Deep Well #15	Ground water	8.55	0.00	0.00	1,258.61	6,615,000.00 *gal*
Application event totals		8.55	0.00	0.00	1,258.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Deep Well #15	Ground water	8.41	0.00	0.00	1,238.63	6,510,000.00 *gal*
Application event totals		8.41	0.00	0.00	1,238.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Deep Well #15	Ground water	8.27	0.00	0.00	1,218.65	6,405,000.00 *gal*
Application event totals		8.27	0.00	0.00	1,218.65	

Dondero 25-4 - 10/20/2022: Rye Grass Silage

Field name: Dondero 25-4

Crop: Rye Grass Silage

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Compost	Compost	242.11	93.10	526.66	6,414.72	600.00 *ton*
Application event totals		242.11	93.10	526.66	6,414.72	

Dondero 25-4 - 06/02/2023: Corn, silage

Field name: Dondero 25-4

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	262.75	115.67	481.24	8,739.36	700.00 *ton*
Application event totals		262.75	115.67	481.24	8,739.36	

06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.99	0.00	0.00	1,127.18	8,442,000.00 *gal*
Application event totals		6.99	0.00	0.00	1,127.18	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.78	0.00	0.00	1,093.53	8,190,000.00 *gal*
Application event totals		6.78	0.00	0.00	1,093.53	

07/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.67	0.00	0.00	1,076.71	8,064,000.00 *gal*
Application event totals		6.67	0.00	0.00	1,076.71	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.52	0.00	0.00	1,051.47	7,875,000.00 *gal*
Application event totals		6.52	0.00	0.00	1,051.47	

Dondero 25-4 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.46	0.00	0.00	1,043.06	7,812,000.00 *gal*
Application event totals		6.46	0.00	0.00	1,043.06	

08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.36	0.00	0.00	1,026.23	7,686,000.00 *gal*
Application event totals		6.36	0.00	0.00	1,026.23	

Field 1 - 10/08/2022: Rye Grass Silage

Field name: Field 1

Crop: Rye Grass Silage Plant date: 10/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	163.02	25.42	161.73	792.06	1,478,900.00 *gal*
Application event totals		163.02	25.42	161.73	792.06	

02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	130.45	20.34	129.41	633.80	1,183,410.00 *gal*
Application event totals		130.45	20.34	129.41	633.80	

Field 1 - 05/28/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage Plant date: 05/28/2023

Field 1 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.78	0.00	0.00	1,094.07	9,150,000.00 *gal*
Application event totals		6.78	0.00	0.00	1,094.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	75.02	10.60	89.48	419.86	2,763,080.00 *gal*
District H20	Ground water	4.11	0.00	0.00	663.61	5,550,000.00 *gal*
Application event totals		79.14	10.60	89.48	1,083.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	4.17	0.00	0.00	672.58	5,625,000.00 *gal*
Application event totals		4.17	0.00	0.00	672.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.97	10.31	87.03	408.37	2,687,460.00 *gal*
Application event totals		72.97	10.31	87.03	408.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.74	10.98	92.72	435.06	2,863,120.00 *gal*
District H20	Ground water	4.00	0.00	0.00	645.68	5,400,000.00 *gal*
Application event totals		81.74	10.98	92.72	1,080.74	

Field 1 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.56	0.00	0.00	1,058.20	8,850,000.00 *gal*
Application event totals		6.56	0.00	0.00	1,058.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.39	0.00	0.00	1,031.29	8,625,000.00 *gal*
Application event totals		6.39	0.00	0.00	1,031.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.28	0.00	0.00	1,013.36	8,475,000.00 *gal*
Application event totals		6.28	0.00	0.00	1,013.36	

Field 1S - 10/17/2022: Rye Grass Silage

Field name: Field 1S

Crop: Rye Grass Silage

Plant date: 10/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	258.21	172.14	153.43	7,409.37	1,200.00 *ton*
Application event totals		258.21	172.14	153.43	7,409.37	

Field 1S - 06/03/2023: Corn, silage

Field name: Field 1S

Crop: Corn, silage

Plant date: 06/03/2023

Field 1S - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	251.89	110.88	461.34	8,377.96	850.00 *ton*
Application event totals		251.89	110.88	461.34	8,377.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.94	0.00	0.00	1,119.46	10,620,000.00 *gal*
Application event totals		6.94	0.00	0.00	1,119.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.59	0.00	0.00	1,062.54	10,080,000.00 *gal*
Application event totals		6.59	0.00	0.00	1,062.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.70	0.00	0.00	1,081.51	10,260,000.00 *gal*
Application event totals		6.70	0.00	0.00	1,081.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.53	0.00	0.00	1,053.05	9,990,000.00 *gal*
Application event totals		6.53	0.00	0.00	1,053.05	

Field 1S - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.53	0.00	0.00	1,053.05	9,990,000.00 *gal*
Application event totals		6.53	0.00	0.00	1,053.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.47	0.00	0.00	1,043.56	9,900,000.00 *gal*
Application event totals		6.47	0.00	0.00	1,043.56	

Field 2 - 10/10/2022: Rye Grass Silage

Field name: Field 2

Crop: Rye Grass Silage Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	355.47	55.43	352.66	1,727.16	1,732,770.00 *gal*
Application event totals		355.47	55.43	352.66	1,727.16	

Field 2 - 05/28/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage Plant date: 05/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 2 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.83	0.00	0.00	1,101.54	4,950,000.00 *gal*
Application event totals		6.83	0.00	0.00	1,101.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.52	0.00	0.00	1,051.47	4,725,000.00 *gal*
Application event totals		6.52	0.00	0.00	1,051.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.43	11.36	95.92	450.10	1,591,560.00 *gal*
District H20	Ground water	4.24	0.00	0.00	684.29	3,075,000.00 *gal*
Application event totals		84.67	11.36	95.92	1,134.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.92	11.01	92.93	436.05	1,541,880.00 *gal*
District H20	Ground water	4.34	0.00	0.00	700.98	3,150,000.00 *gal*
Application event totals		82.26	11.01	92.93	1,137.03	

Field 2 - 05/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	84.36	11.92	100.61	472.09	1,669,320.00 *gal*
District H20	Ground water	4.14	0.00	0.00	667.60	3,000,000.00 *gal*
Application event totals		88.49	11.92	100.61	1,139.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.41	0.00	0.00	1,034.78	4,650,000.00 *gal*
Application event totals		6.41	0.00	0.00	1,034.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.21	0.00	0.00	1,001.40	4,500,000.00 *gal*
Application event totals		6.21	0.00	0.00	1,001.40	

Field 3 - 10/08/2022: Rye Grass Silage

Field name: Field 3

Crop: Rye Grass Silage

Plant date: 10/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	168.56	26.31	195.66	1,271.39	2,341,500.00 *gal*
Application event totals		168.56	26.31	195.66	1,271.39	

Field 3 - 10/08/2022: Rye Grass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.61	9.31	69.19	449.59	828,000.00 *gal*
IW 10	Ground water	0.04	0.00	0.00	3.42	162,000.00 *gal*
Application event totals		59.65	9.31	69.19	453.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	336.92	52.54	334.25	1,637.01	3,421,500.00 *gal*
Application event totals		336.92	52.54	334.25	1,637.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	38.40	0.00	0.00	0.00	
District H20	Ground water	5.46	0.00	0.00	881.23	8,250,000.00 *gal*
Application event totals		43.86	0.00	0.00	881.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 10	Ground water	1.22	0.00	0.00	97.04	4,590,000.00 *gal*
Application event totals		1.22	0.00	0.00	97.04	

Field 3 - 06/14/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 3 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.65	0.00	0.00	1,073.50	10,050,000.00 *gal*
Application event totals		6.65	0.00	0.00	1,073.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.45	0.00	0.00	1,041.46	9,750,000.00 *gal*
Application event totals		6.45	0.00	0.00	1,041.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	85.30	12.05	101.74	477.38	3,516,750.00 *gal*
District H20	Ground water	4.07	0.00	0.00	656.92	6,150,000.00 *gal*
Application event totals		89.37	12.05	101.74	1,134.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.55	11.10	93.69	439.61	3,238,500.00 *gal*
District H20	Ground water	4.12	0.00	0.00	664.93	6,225,000.00 *gal*
Application event totals		82.67	11.10	93.69	1,104.54	

Field 3 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	76.33	10.78	91.04	427.19	3,147,000.00 *gal*
District H20	Ground water	4.22	0.00	0.00	680.95	6,375,000.00 *gal*
Application event totals		80.55	10.78	91.04	1,108.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.36	0.00	0.00	1,025.43	9,600,000.00 *gal*
Application event totals		6.36	0.00	0.00	1,025.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.31	0.00	0.00	1,017.42	9,525,000.00 *gal*
Application event totals		6.31	0.00	0.00	1,017.42	

Field 3S - 10/18/2022: Rye Grass Silage

Field name: Field 3S

Crop: Rye Grass Silage Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	262.27	174.84	155.84	7,525.87	850.00 *ton*
Application event totals		262.27	174.84	155.84	7,525.87	

Field 3S - 06/01/2023: Corn, silage

Field name: Field 3S

Crop: Corn, silage Plant date: 06/01/2023

Field 3S - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	254.96	112.24	466.97	8,480.24	600.00 *ton*
Application event totals		254.96	112.24	466.97	8,480.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	7.17	0.00	0.00	1,156.33	7,650,000.00 *gal*
Application event totals		7.17	0.00	0.00	1,156.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.75	0.00	0.00	1,088.31	7,200,000.00 *gal*
Application event totals		6.75	0.00	0.00	1,088.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.75	0.00	0.00	1,088.31	7,200,000.00 *gal*
Application event totals		6.75	0.00	0.00	1,088.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.58	0.00	0.00	1,061.11	7,020,000.00 *gal*
Application event totals		6.58	0.00	0.00	1,061.11	

Field 3S - 06/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.41	0.00	0.00	1,033.90	6,840,000.00 *gal*
Application event totals		6.41	0.00	0.00	1,033.90	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.32	0.00	0.00	1,020.29	6,750,000.00 *gal*
Application event totals		6.32	0.00	0.00	1,020.29	

Field 4 - 10/12/2022: Rye Grass Silage

Field name: Field 4

Crop: Rye Grass Silage

Plant date: 10/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	313.36	48.86	310.88	1,522.54	3,733,840.00 *gal*
Application event totals		313.36	48.86	310.88	1,522.54	

05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	214.36	20.76	262.03	1,094.97	2,872,320.00 *gal*
District H20	Ground water	3.94	0.00	0.00	634.98	6,975,000.00 *gal*
Application event totals		218.30	20.76	262.03	1,729.95	

Field 4 - 10/12/2022: Rye Grass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Solid commercial fertilizer	38.40	0.00	0.00	0.00	
District H20	Ground water	4.99	0.00	0.00	805.67	8,850,000.00 *gal*
Application event totals		43.39	0.00	0.00	805.67	

Field 4 - 06/23/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.77	0.00	0.00	1,092.44	12,000,000.00 *gal*
Application event totals		6.77	0.00	0.00	1,092.44	

07/29/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.43	0.00	0.00	1,037.81	11,400,000.00 *gal*
Application event totals		6.43	0.00	0.00	1,037.81	

Field 4 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.34	11.07	93.43	438.39	3,789,280.00 *gal*
District H20	Ground water	4.23	0.00	0.00	682.77	7,500,000.00 *gal*
Application event totals		82.57	11.07	93.43	1,121.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	75.99	10.74	90.63	425.26	3,675,760.00 *gal*
District H20	Ground water	4.15	0.00	0.00	669.12	7,350,000.00 *gal*
Application event totals		80.14	10.74	90.63	1,094.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.73	10.28	86.74	407.03	3,518,240.00 *gal*
District H20	Ground water	4.27	0.00	0.00	689.60	7,575,000.00 *gal*
Application event totals		77.01	10.28	86.74	1,096.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.22	0.00	0.00	1,003.68	11,025,000.00 *gal*
Application event totals		6.22	0.00	0.00	1,003.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.14	0.00	0.00	990.02	10,875,000.00 *gal*
Application event totals		6.14	0.00	0.00	990.02	

Field 5 - 10/15/2022: Rye Grass Silage

Field name: Field 5

Crop: Rye Grass Silage

Plant date: 10/15/2022

Field 5 - 10/15/2022: Rye Grass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/27/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	190.27	29.70	220.85	1,435.10	1,762,000.00 *gal*
IW 10	Ground water	1.62	0.00	0.00	128.43	4,050,000.00 *gal*
Application event totals		191.88	29.70	220.85	1,563.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	236.50	22.90	289.08	1,208.02	1,800,500.00 *gal*
Application event totals		236.50	22.90	289.08	1,208.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	5.51	0.00	0.00	889.24	5,550,000.00 *gal*
Application event totals		5.51	0.00	0.00	889.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	38.40	0.00	0.00	0.00	
Application event totals		38.40	0.00	0.00	0.00	

Field 5 - 06/10/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.70	0.00	0.00	1,081.51	6,750,000.00 *gal*
Application event totals		6.70	0.00	0.00	1,081.51	

Field 5 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.48	0.00	0.00	1,045.46	6,525,000.00 *gal*
Application event totals		6.48	0.00	0.00	1,045.46	

07/10/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.25	11.34	95.71	449.08	2,205,500.00 *gal*
District H20	Ground water	4.17	0.00	0.00	672.94	4,200,000.00 *gal*
Application event totals		84.42	11.34	95.71	1,122.02	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.01	11.02	93.04	436.56	2,144,000.00 *gal*
District H20	Ground water	4.02	0.00	0.00	648.91	4,050,000.00 *gal*
Application event totals		82.03	11.02	93.04	1,085.46	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	73.30	10.36	87.42	410.19	2,014,500.00 *gal*
District H20	Ground water	4.02	0.00	0.00	648.91	4,050,000.00 *gal*
Application event totals		77.32	10.36	87.42	1,059.10	

Field 5 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.33	0.00	0.00	1,021.43	6,375,000.00 *gal*
Application event totals		6.33	0.00	0.00	1,021.43	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.33	0.00	0.00	1,021.43	6,375,000.00 *gal*
Application event totals		6.33	0.00	0.00	1,021.43	

Field 6 - 10/08/2022: Rye Grass Silage

Field name: Field 6

Crop: Rye Grass Silage Plant date: 10/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	186.16	29.06	216.09	1,404.15	1,724,000.00 *gal*
IW 10	Ground water	1.58	0.00	0.00	125.58	3,960,000.00 *gal*
Application event totals		187.74	29.06	216.09	1,529.73	

04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	238.60	23.10	291.65	1,218.76	1,816,500.00 *gal*
District H20	Ground water	3.72	0.00	0.00	600.84	3,750,000.00 *gal*
Application event totals		242.32	23.10	291.65	1,819.60	

Field 6 - 10/08/2022: Rye Grass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	5.21	0.00	0.00	841.18	5,250,000.00 *gal*
Application event totals		5.21	0.00	0.00	841.18	

05/19/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	38.40	0.00	0.00	0.00	
Application event totals		38.40	0.00	0.00	0.00	

Field 6 - 06/16/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.70	0.00	0.00	1,081.51	6,750,000.00 *gal*
Application event totals		6.70	0.00	0.00	1,081.51	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.48	0.00	0.00	1,045.46	6,525,000.00 *gal*
Application event totals		6.48	0.00	0.00	1,045.46	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Field 6 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	77.79	10.99	92.78	435.34	2,138,000.00 *gal*
District H20	Ground water	4.17	0.00	0.00	672.94	4,200,000.00 *gal*
Application event totals		81.96	10.99	92.78	1,108.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.37	10.51	88.70	416.20	2,044,000.00 *gal*
District H20	Ground water	4.10	0.00	0.00	660.92	4,125,000.00 *gal*
Application event totals		78.47	10.51	88.70	1,077.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.44	10.09	85.20	399.80	1,963,500.00 *gal*
District H20	Ground water	4.10	0.00	0.00	660.92	4,125,000.00 *gal*
Application event totals		75.54	10.09	85.20	1,060.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.33	0.00	0.00	1,021.43	6,375,000.00 *gal*
Application event totals		6.33	0.00	0.00	1,021.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.33	0.00	0.00	1,021.43	6,375,000.00 *gal*
Application event totals		6.33	0.00	0.00	1,021.43	

Field 7 - 10/08/2022: Rye Grass Silage

Field name: Field 7

Crop: Rye Grass Silage

Plant date: 10/08/2022

Field 7 - 10/08/2022: Rye Grass Silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	196.30	19.01	239.95	1,002.72	2,181,970.00 *gal*
District H20	Ground water	3.98	0.00	0.00	641.99	5,850,000.00 *gal*
Application event totals		200.28	19.01	239.95	1,644.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	5.36	0.00	0.00	864.22	7,875,000.00 *gal*
Application event totals		5.36	0.00	0.00	864.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	38.40	0.00	0.00	0.00	
Application event totals		38.40	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	183.41	28.63	212.89	1,383.38	2,479,810.00 *gal*
IW 13	Ground water	1.41	0.00	0.00	140.61	6,000,000.00 *gal*
Application event totals		184.82	28.63	212.89	1,523.99	

Field 7 - 06/17/2023: Corn, silage

Field name: Field 7

Crop: Corn, silage

Plant date: 06/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 7 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.78	0.00	0.00	1,094.68	9,975,000.00 *gal*
Application event totals		6.78	0.00	0.00	1,094.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.48	0.00	0.00	1,045.30	9,525,000.00 *gal*
Application event totals		6.48	0.00	0.00	1,045.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.43	0.00	0.00	1,037.07	9,450,000.00 *gal*
Application event totals		6.43	0.00	0.00	1,037.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
District H20	Ground water	6.33	0.00	0.00	1,020.60	9,300,000.00 *gal*
Application event totals		6.33	0.00	0.00	1,020.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.13	9.91	83.64	392.47	2,814,150.00 *gal*
District H20	Ground water	4.49	0.00	0.00	724.30	6,600,000.00 *gal*
Application event totals		74.62	9.91	83.64	1,116.77	

Field 7 - 06/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.11	9.76	82.43	386.77	2,773,270.00 *gal*
District H20	Ground water	4.39	0.00	0.00	707.84	6,450,000.00 *gal*
Application event totals		73.50	9.76	82.43	1,094.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.18	9.63	81.32	381.58	2,736,040.00 *gal*
District H20	Ground water	4.39	0.00	0.00	707.84	6,450,000.00 *gal*
Application event totals		72.57	9.63	81.32	1,089.42	

Field 8 - 10/16/2022: Rye Grass Silage

Field name: Field 8

Crop: Rye Grass Silage Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	256.20	170.80	152.23	7,351.74	940.00 *ton*
Application event totals		256.20	170.80	152.23	7,351.74	

Field 8 - 06/04/2023: Corn, silage

Field name: Field 8

Crop: Corn, silage Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 8 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	247.74	109.06	453.74	8,239.97	550.00 *ton*
Application event totals		247.74	109.06	453.74	8,239.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Deep Well #10	Ground water	2.44	0.00	0.00	1,285.80	7,200,000.00 *gal*
Application event totals		2.44	0.00	0.00	1,285.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Deep Well #10	Ground water	2.32	0.00	0.00	1,221.51	6,840,000.00 *gal*
Application event totals		2.32	0.00	0.00	1,221.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Deep Well #10	Ground water	2.26	0.00	0.00	1,189.36	6,660,000.00 *gal*
Application event totals		2.26	0.00	0.00	1,189.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Deep Well #10	Ground water	2.20	0.00	0.00	1,157.22	6,480,000.00 *gal*
Application event totals		2.20	0.00	0.00	1,157.22	

Field 8 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Deep Well #10	Ground water	2.20	0.00	0.00	1,157.22	6,480,000.00 *gal*
Application event totals		2.20	0.00	0.00	1,157.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Deep Well #10	Ground water	2.13	0.00	0.00	1,125.07	6,300,000.00 *gal*
Application event totals		2.13	0.00	0.00	1,125.07	

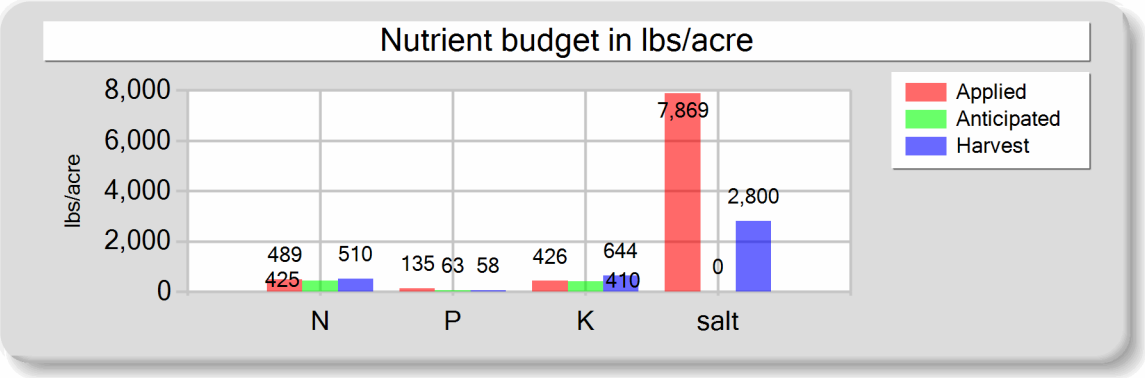
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Deep Well #10	Ground water	2.13	0.00	0.00	1,125.07	6,300,000.00 *gal*
Application event totals		2.13	0.00	0.00	1,125.07	

B. NUTRIENT BUDGET

Dondero 25-14 - 10/08/2022: Rye Grass Silage

Field name: Dondero 25-14 Crop: Rye Grass Silage Plant date: 10/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	38.40	0.00	0.00	0.00
Dry manure	163.53	109.02	97.17	4,692.60
Process wastewater	268.87	26.03	328.66	1,373.41
Fresh water	11.17	0.00	0.00	1,802.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	488.98	135.05	425.83	7,868.53
Anticipated crop nutrient removal	425.00	62.50	410.00	0.00
Actual crop nutrient removal	509.73	58.28	643.68	2,800.02
Nutrient balance	-20.75	76.77	-217.85	5,068.51
Applied to removed ratio	0.96	2.32	0.66	2.81

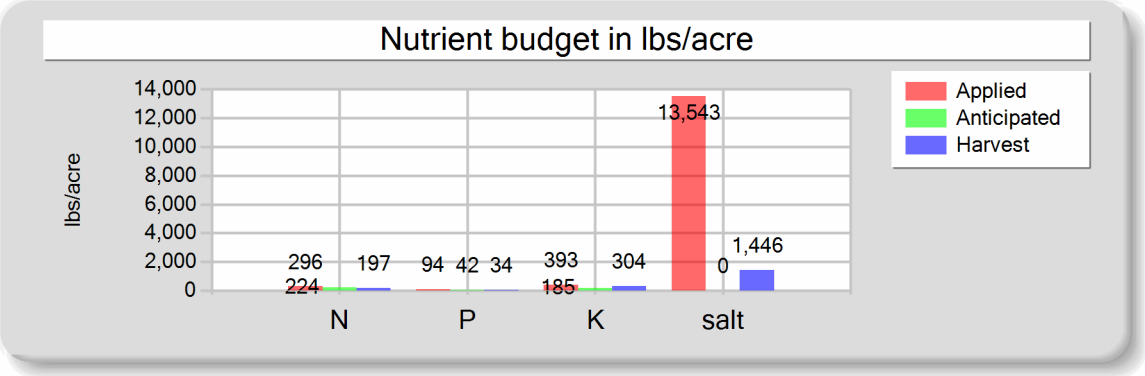
Fresh water applied	
14,175,000.00	*gallons*
522.02	*acre-inches*
8.29	*inches/acre*

Process wastewater applied	
2,579,220.00	*gallons*
94.98	*acre-inches*
1.51	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dondero 25-14 - 06/27/2023: Corn, silage

Field name: Dondero 25-14 Crop: Corn, silage Plant date: 06/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	214.49	94.42	392.85	7,134.17
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	39.72	0.00	0.00	6,408.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	296.41	94.42	392.85	13,543.13
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	197.24	34.38	304.00	1,445.81
Nutrient balance	99.18	60.04	88.85	12,097.32
Applied to removed ratio	1.50	2.75	1.29	9.37

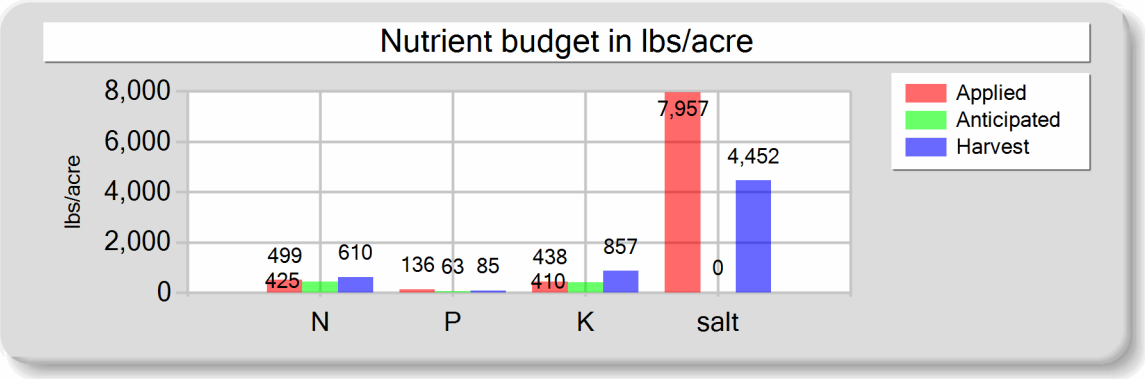
Fresh water applied	
50,400,000.00	*gallons*
1,856.06	*acre-inches*
29.46	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dondero 25-15 - 10/08/2022: Rye Grass Silage

Field name: Dondero 25-15 Crop: Rye Grass Silage Plant date: 10/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	38.40	0.00	0.00	0.00
Dry manure	163.53	109.02	97.17	4,692.60
Process wastewater	278.66	26.98	340.62	1,423.39
Fresh water	11.41	0.00	0.00	1,840.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	499.00	136.00	437.79	7,956.66
Anticipated crop nutrient removal	425.00	62.50	410.00	0.00
Actual crop nutrient removal	609.79	85.49	856.68	4,451.76
Nutrient balance	-110.80	50.51	-418.88	3,504.90
Applied to removed ratio	0.82	1.59	0.51	1.79

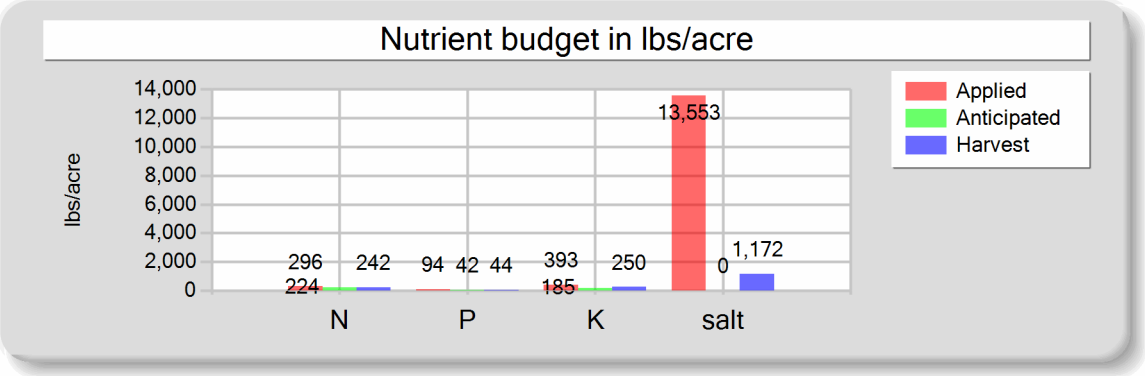
Fresh water applied	
14,475,000.00	*gallons*
533.07	*acre-inches*
8.46	*inches/acre*

Process wastewater applied	
2,673,090.00	*gallons*
98.44	*acre-inches*
1.56	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dondero 25-15 - 07/03/2023: Corn, silage

Field name: Dondero 25-15 Crop: Corn, silage Plant date: 07/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	214.49	94.42	392.85	7,134.17
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	39.78	0.00	0.00	6,418.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	296.47	94.42	392.85	13,552.67
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	241.71	44.36	250.07	1,171.80
Nutrient balance	54.76	50.07	142.77	12,380.87
Applied to removed ratio	1.23	2.13	1.57	11.57

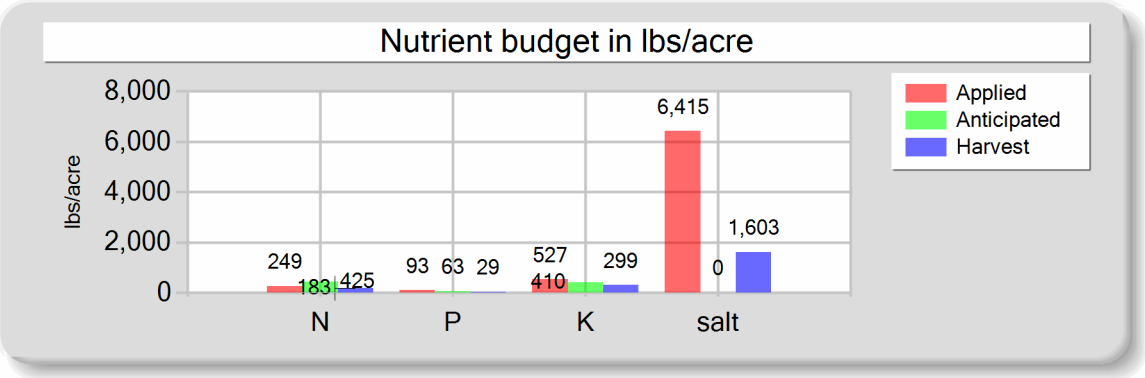
Fresh water applied	
50,475,000.00	*gallons*
1,858.82	*acre-inches*
29.51	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dondero 25-19 - 10/22/2022: Rye Grass Silage

Field name: Dondero 25-19 Crop: Rye Grass Silage Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	242.11	93.10	526.66	6,414.72
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	249.11	93.10	526.66	6,414.72
Anticipated crop nutrient removal	425.00	62.50	410.00	0.00
Actual crop nutrient removal	183.49	29.28	298.66	1,603.37
Nutrient balance	65.62	63.82	228.00	4,811.35
Applied to removed ratio	1.36	3.18	1.76	4.00

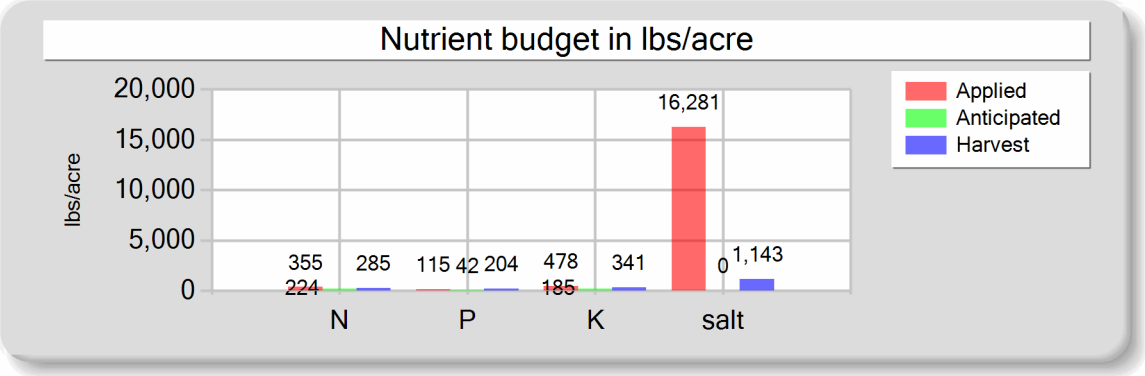
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Dondero 25-19 - 06/09/2023: Corn, silage

Field name: Dondero 25-19 Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	261.25	115.01	478.49	8,689.42
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	51.54	0.00	0.00	7,591.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	354.99	115.01	478.49	16,281.03
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	284.64	203.85	341.26	1,142.62
Nutrient balance	70.36	-88.84	137.23	15,138.42
Applied to removed ratio	1.25	0.56	1.40	14.25

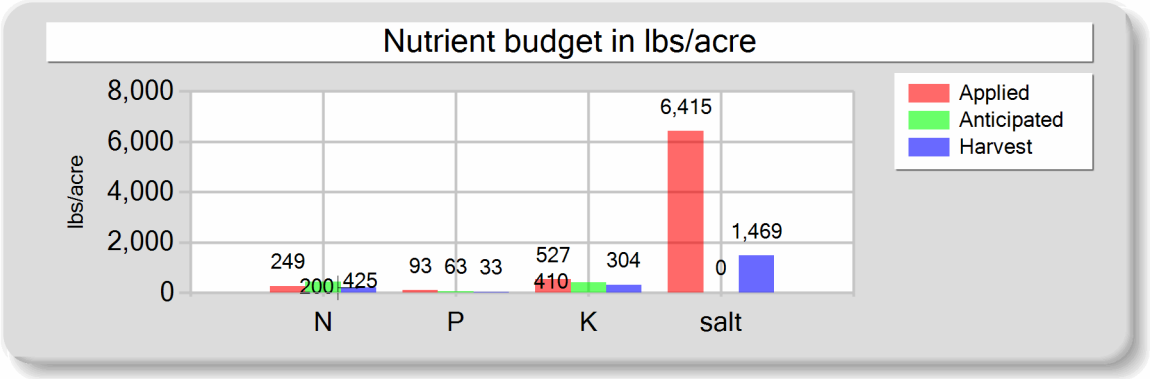
Fresh water applied	
39,900,000.00	*gallons*
1,469.38	*acre-inches*
29.39	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dondero 25-4 - 10/20/2022: Rye Grass Silage

Field name: Dondero 25-4 Crop: Rye Grass Silage Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	242.11	93.10	526.66	6,414.72
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	249.11	93.10	526.66	6,414.72
Anticipated crop nutrient removal	425.00	62.50	410.00	0.00
Actual crop nutrient removal	200.18	32.97	303.80	1,468.95
Nutrient balance	48.94	60.13	222.86	4,945.77
Applied to removed ratio	1.24	2.82	1.73	4.37

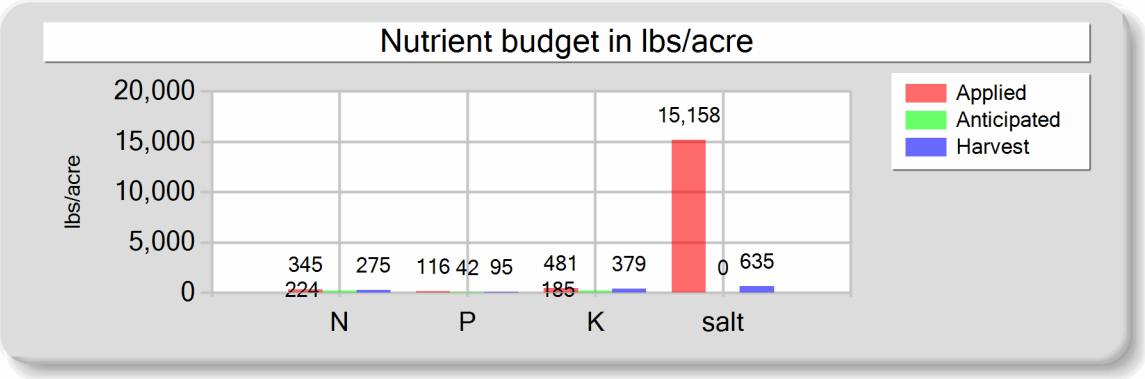
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Dondero 25-4 - 06/02/2023: Corn, silage

Field name: Dondero 25-4 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	262.75	115.67	481.24	8,739.36
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	39.78	0.00	0.00	6,418.17
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	344.73	115.67	481.24	15,157.53
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	274.85	94.63	378.53	635.02
Nutrient balance	69.88	21.03	102.70	14,522.52
Applied to removed ratio	1.25	1.22	1.27	23.87

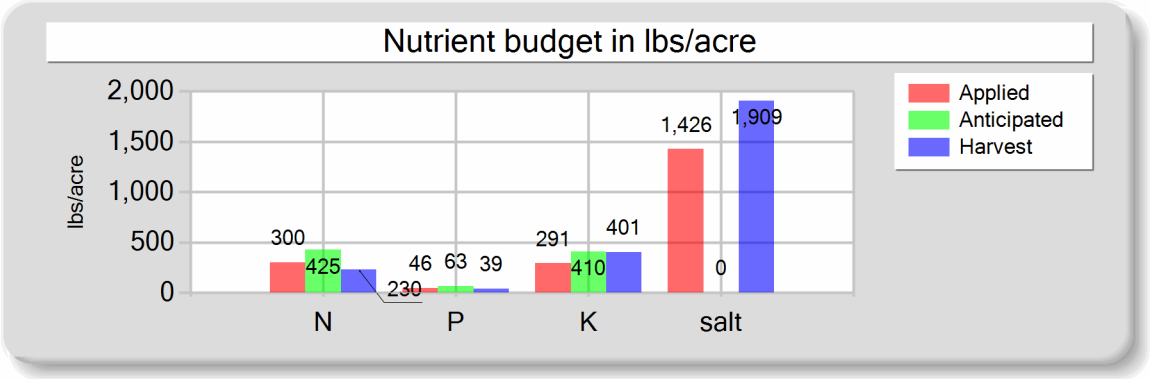
Fresh water applied	
48,069,000.00	*gallons*
1,770.22	*acre-inches*
29.50	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 10/08/2022: Rye Grass Silage

Field name: Field 1 Crop: Rye Grass Silage Plant date: 10/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	293.46	45.76	291.14	1,425.87
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	300.46	45.76	291.14	1,425.87
Anticipated crop nutrient removal	425.00	62.50	410.00	0.00
Actual crop nutrient removal	230.27	39.29	400.52	1,908.76
Nutrient balance	70.19	6.47	-109.38	-482.89
Applied to removed ratio	1.30	1.16	0.73	0.75

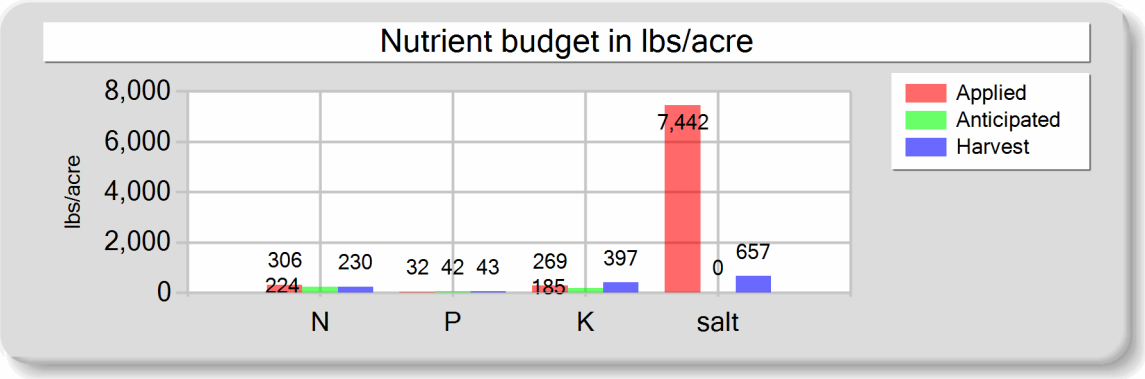
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
2,662,310.00 *gallons*
98.04 *acre-inches*
1.46 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 1 - 05/28/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	225.74	31.89	269.23	1,263.29
Fresh water	38.30	0.00	0.00	6,178.79
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	306.23	31.89	269.23	7,442.08
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	229.59	43.01	397.20	656.51
Nutrient balance	76.64	-11.12	-127.97	6,785.57
Applied to removed ratio	1.33	0.74	0.68	11.34

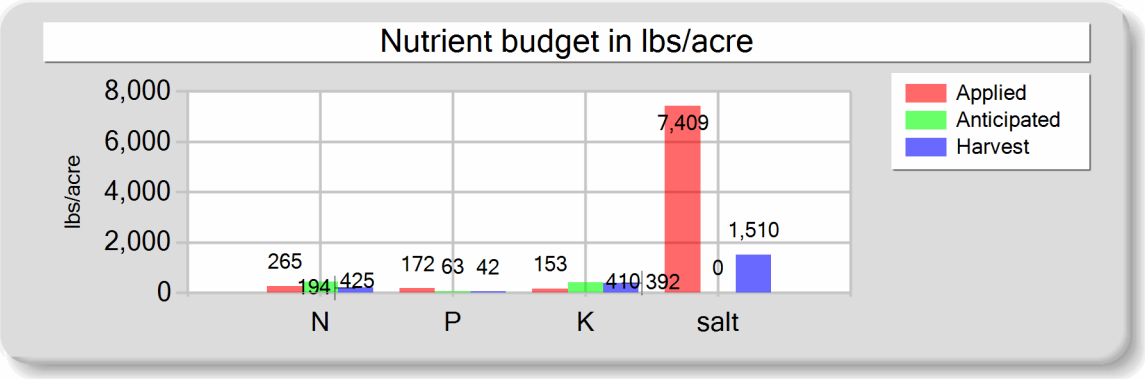
Fresh water applied	
51,675,000.00	*gallons*
1,903.01	*acre-inches*
28.40	*inches/acre*

Process wastewater applied	
8,313,660.00	*gallons*
306.16	*acre-inches*
4.57	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1S - 10/17/2022: Rye Grass Silage

Field name: Field 1S Crop: Rye Grass Silage Plant date: 10/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	258.21	172.14	153.43	7,409.37
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	265.21	172.14	153.43	7,409.37
Anticipated crop nutrient removal	425.00	62.50	410.00	0.00
Actual crop nutrient removal	194.03	42.26	391.89	1,509.51
Nutrient balance	71.18	129.87	-238.47	5,899.86
Applied to removed ratio	1.37	4.07	0.39	4.91

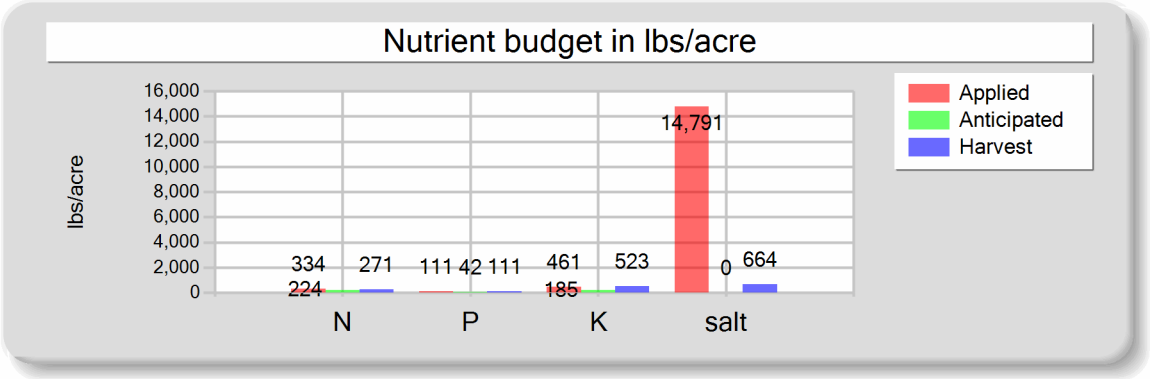
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 1S - 06/03/2023: Corn, silage

Field name: Field 1S Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	251.89	110.88	461.34	8,377.96
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	39.75	0.00	0.00	6,413.18
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	333.83	110.88	461.34	14,791.13
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	271.06	110.51	523.44	663.98
Nutrient balance	62.78	0.37	-62.11	14,127.16
Applied to removed ratio	1.23	1.00	0.88	22.28

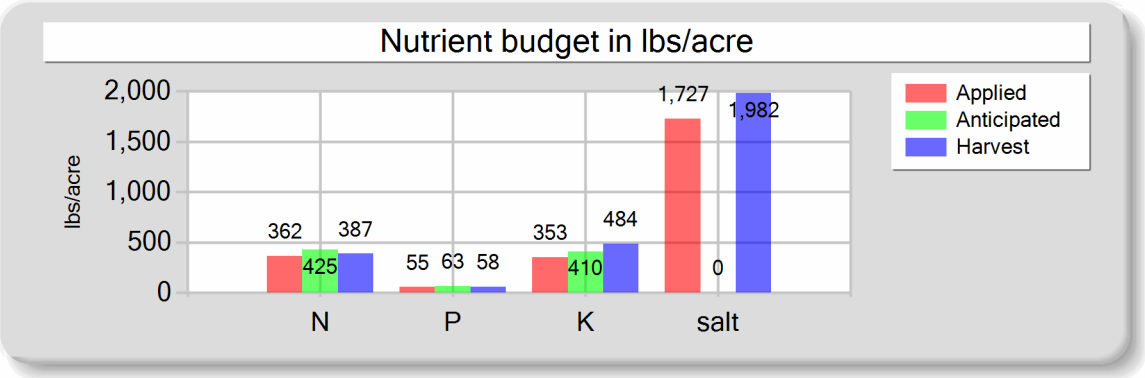
Fresh water applied	
60,840,000.00	*gallons*
2,240.53	*acre-inches*
29.48	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 10/10/2022: Rye Grass Silage

Field name: Field 2 Crop: Rye Grass Silage Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	355.47	55.43	352.66	1,727.16
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	362.47	55.43	352.66	1,727.16
Anticipated crop nutrient removal	425.00	62.50	410.00	0.00
Actual crop nutrient removal	387.17	58.16	484.10	1,982.04
Nutrient balance	-24.69	-2.73	-131.44	-254.87
Applied to removed ratio	0.94	0.95	0.73	0.87

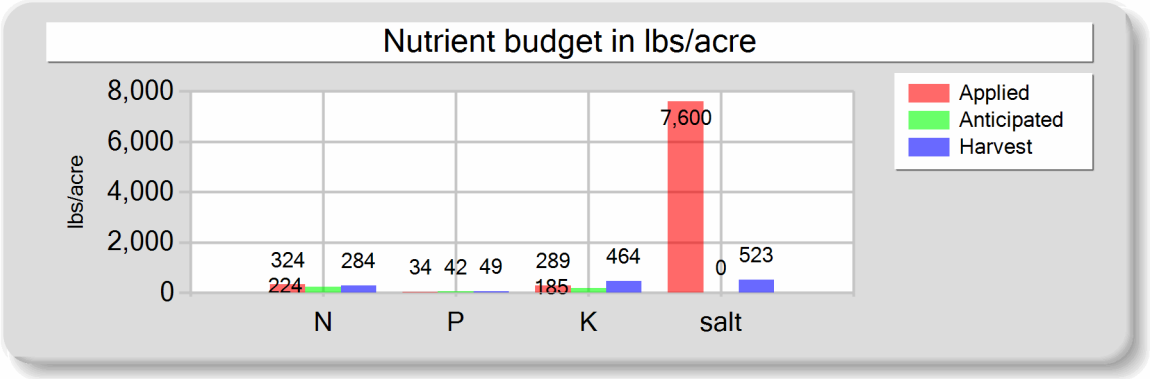
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,732,770.00	*gallons*
63.81	*acre-inches*
1.77	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 05/28/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	242.70	34.29	289.46	1,358.23
Fresh water	38.69	0.00	0.00	6,242.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	323.59	34.29	289.46	7,600.29
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	283.97	48.64	463.50	522.73
Nutrient balance	39.62	-14.35	-174.04	7,077.57
Applied to removed ratio	1.14	0.70	0.62	14.54

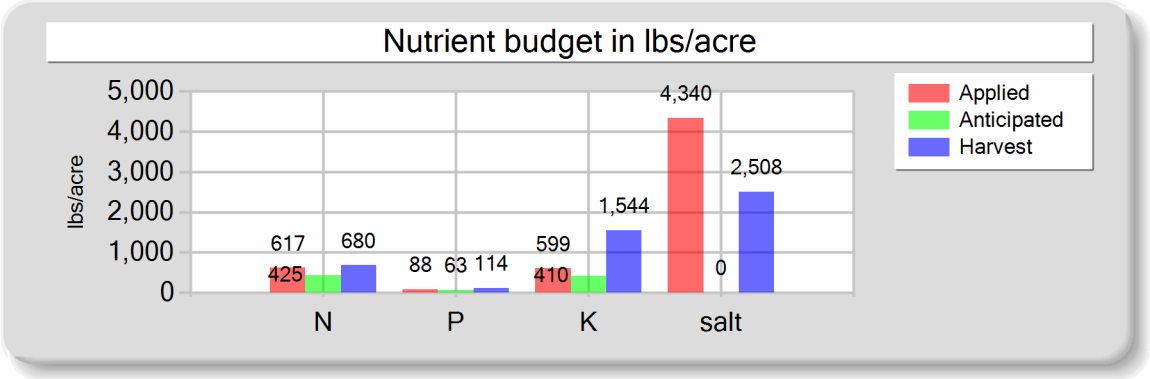
Fresh water applied	
28,050,000.00	*gallons*
1,032.99	*acre-inches*
28.69	*inches/acre*

Process wastewater applied	
4,802,760.00	*gallons*
176.87	*acre-inches*
4.91	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 10/08/2022: Rye Grass Silage

Field name: Field 3 Crop: Rye Grass Silage Plant date: 10/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	38.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	565.09	88.16	599.10	3,357.98
Fresh water	6.73	0.00	0.00	981.69
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	617.21	88.16	599.10	4,339.68
Anticipated crop nutrient removal	425.00	62.50	410.00	0.00
Actual crop nutrient removal	680.37	114.09	1,543.64	2,507.74
Nutrient balance	-63.16	-25.94	-944.54	1,831.94
Applied to removed ratio	0.91	0.77	0.39	1.73

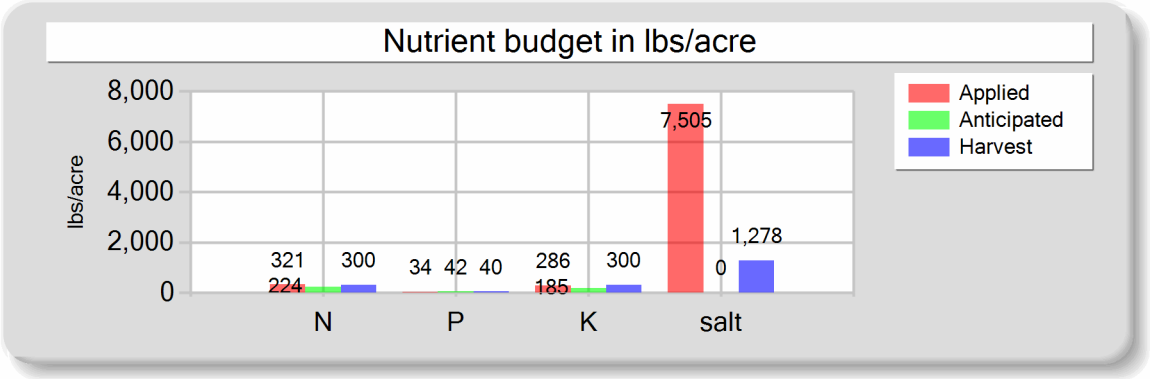
Fresh water applied	
13,002,000.00	*gallons*
478.82	*acre-inches*
6.38	*inches/acre*

Process wastewater applied	
6,591,000.00	*gallons*
242.72	*acre-inches*
3.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 06/14/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	240.19	33.94	286.47	1,344.18
Fresh water	38.18	0.00	0.00	6,160.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	320.57	33.94	286.47	7,504.80
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	300.30	40.30	300.30	1,277.64
Nutrient balance	20.27	-6.36	-13.83	6,227.16
Applied to removed ratio	1.07	0.84	0.95	5.87

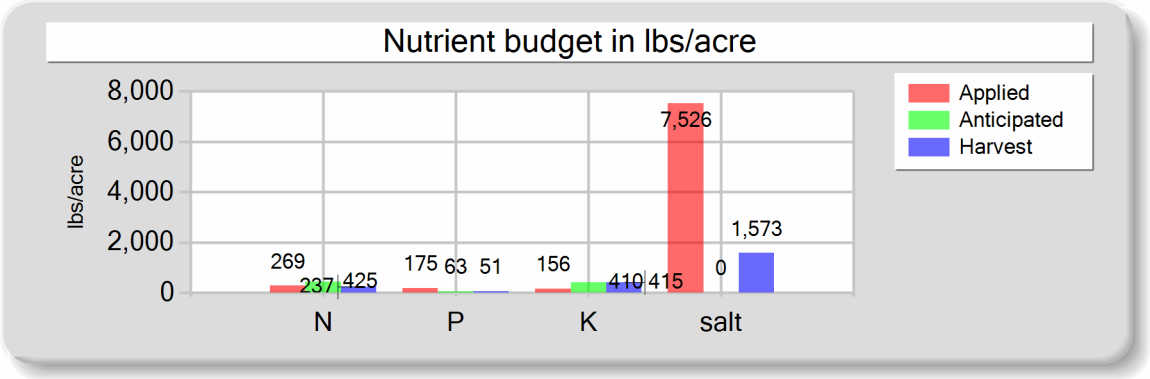
Fresh water applied	
57,675,000.00	*gallons*
2,123.97	*acre-inches*
28.32	*inches/acre*

Process wastewater applied	
9,902,250.00	*gallons*
364.67	*acre-inches*
4.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3S - 10/18/2022: Rye Grass Silage

Field name: Field 3S Crop: Rye Grass Silage Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	262.27	174.84	155.84	7,525.87
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	269.27	174.84	155.84	7,525.87
Anticipated crop nutrient removal	425.00	62.50	410.00	0.00
Actual crop nutrient removal	237.28	50.54	415.45	1,572.72
Nutrient balance	31.99	124.30	-259.61	5,953.14
Applied to removed ratio	1.13	3.46	0.38	4.79

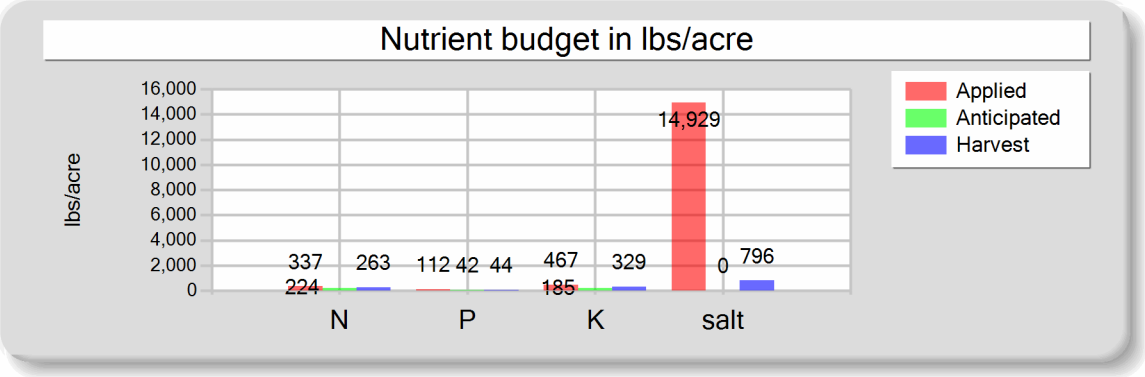
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 3S - 06/01/2023: Corn, silage

Field name: Field 3S Crop: Corn, silage Plant date: 06/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	254.96	112.24	466.97	8,480.24
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	39.97	0.00	0.00	6,448.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	337.13	112.24	466.97	14,928.50
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	262.96	44.31	329.07	795.86
Nutrient balance	74.17	67.93	137.90	14,132.64
Applied to removed ratio	1.28	2.53	1.42	18.76

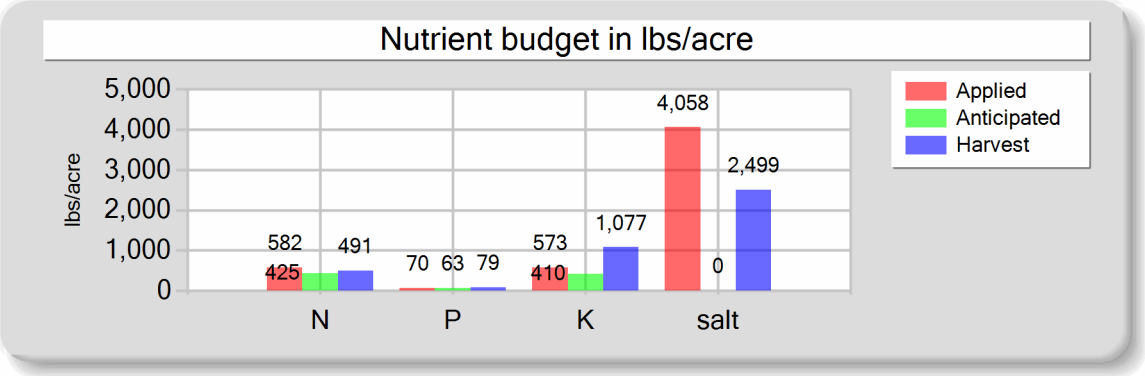
Fresh water applied	
42,660,000.00	*gallons*
1,571.02	*acre-inches*
29.64	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 10/12/2022: Rye Grass Silage

Field name: Field 4 Crop: Rye Grass Silage Plant date: 10/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	38.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	527.72	69.62	572.91	2,617.51
Fresh water	8.93	0.00	0.00	1,440.65
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	582.05	69.62	572.91	4,058.16
Anticipated crop nutrient removal	425.00	62.50	410.00	0.00
Actual crop nutrient removal	490.78	79.05	1,077.11	2,499.33
Nutrient balance	91.27	-9.43	-504.20	1,558.83
Applied to removed ratio	1.19	0.88	0.53	1.62

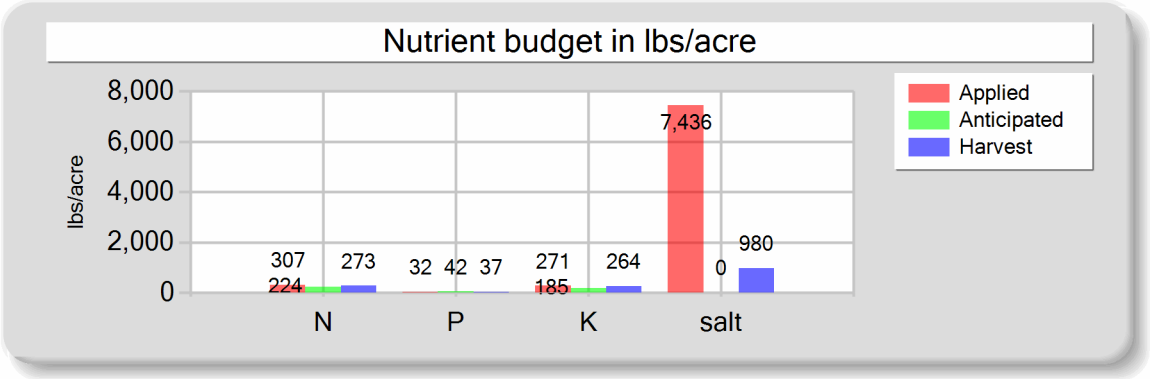
Fresh water applied	
15,825,000.00	*gallons*
582.78	*acre-inches*
6.62	*inches/acre*

Process wastewater applied	
6,606,160.00	*gallons*
243.28	*acre-inches*
2.76	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 06/23/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	227.06	32.08	270.80	1,270.68
Fresh water	38.21	0.00	0.00	6,165.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	307.47	32.08	270.80	7,436.12
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	273.30	37.45	263.94	979.90
Nutrient balance	34.17	-5.38	6.86	6,456.21
Applied to removed ratio	1.13	0.86	1.03	7.59

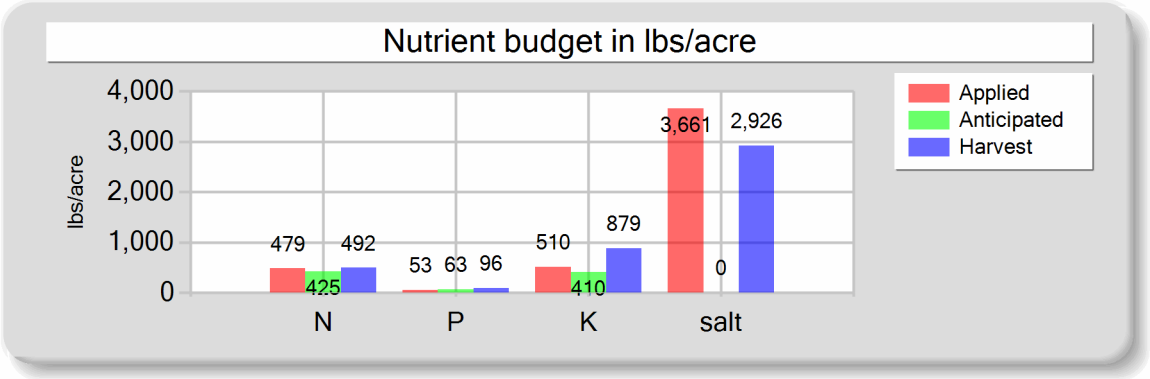
Fresh water applied	
67,725,000.00	*gallons*
2,494.08	*acre-inches*
28.34	*inches/acre*

Process wastewater applied	
10,983,280.00	*gallons*
404.48	*acre-inches*
4.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 10/15/2022: Rye Grass Silage

Field name: Field 5 Crop: Rye Grass Silage Plant date: 10/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	38.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	426.76	52.60	509.94	2,643.12
Fresh water	7.13	0.00	0.00	1,017.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	479.29	52.60	509.94	3,660.80
Anticipated crop nutrient removal	425.00	62.50	410.00	0.00
Actual crop nutrient removal	492.14	96.13	879.11	2,926.21
Nutrient balance	-12.85	-43.53	-369.18	734.59
Applied to removed ratio	0.97	0.55	0.58	1.25

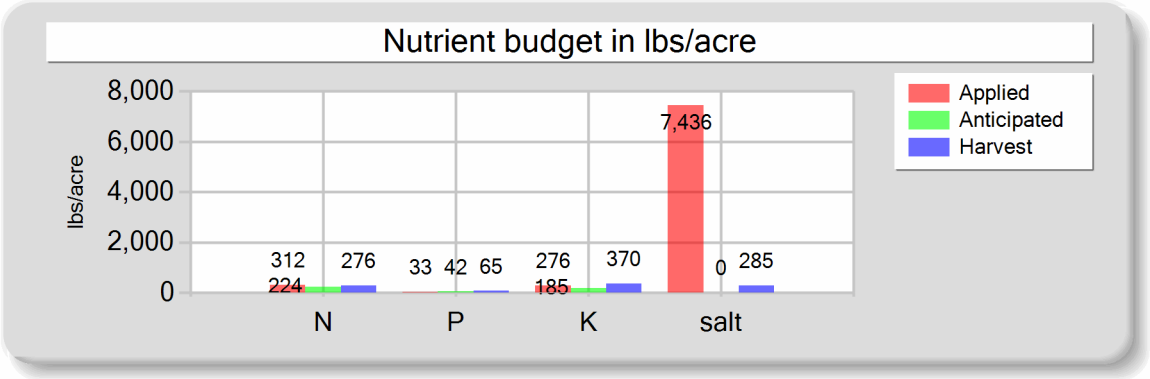
Fresh water applied	
9,600,000.00	*gallons*
353.54	*acre-inches*
7.07	*inches/acre*

Process wastewater applied	
3,562,500.00	*gallons*
131.19	*acre-inches*
2.62	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 06/10/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	231.55	32.71	276.16	1,295.82
Fresh water	38.06	0.00	0.00	6,140.58
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	311.81	32.71	276.16	7,436.41
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	276.32	65.40	369.51	284.82
Nutrient balance	35.49	-32.69	-93.35	7,151.59
Applied to removed ratio	1.13	0.50	0.75	26.11

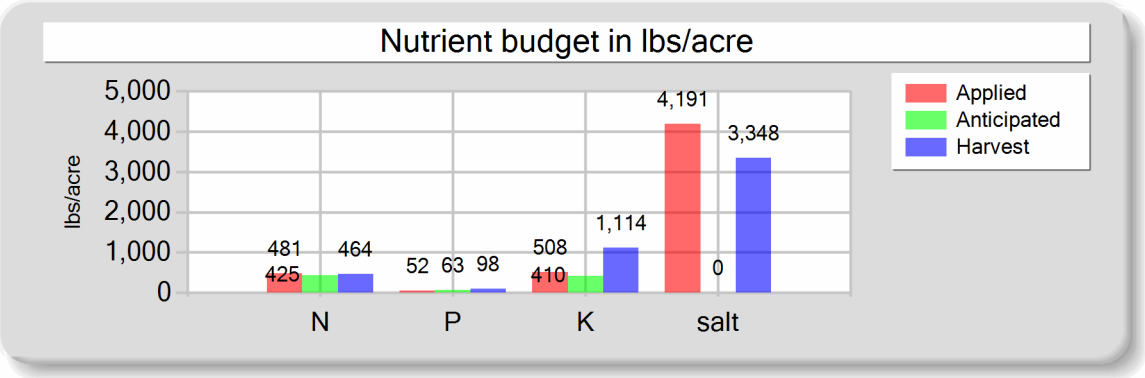
Fresh water applied	
38,325,000.00	*gallons*
1,411.38	*acre-inches*
28.23	*inches/acre*

Process wastewater applied	
6,364,000.00	*gallons*
234.36	*acre-inches*
4.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 10/08/2022: Rye Grass Silage

Field name: Field 6 Crop: Rye Grass Silage Plant date: 10/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	38.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	424.76	52.16	507.74	2,622.91
Fresh water	10.52	0.00	0.00	1,567.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	480.68	52.16	507.74	4,190.50
Anticipated crop nutrient removal	425.00	62.50	410.00	0.00
Actual crop nutrient removal	463.70	97.67	1,113.62	3,348.15
Nutrient balance	16.98	-45.51	-605.87	842.35
Applied to removed ratio	1.04	0.53	0.46	1.25

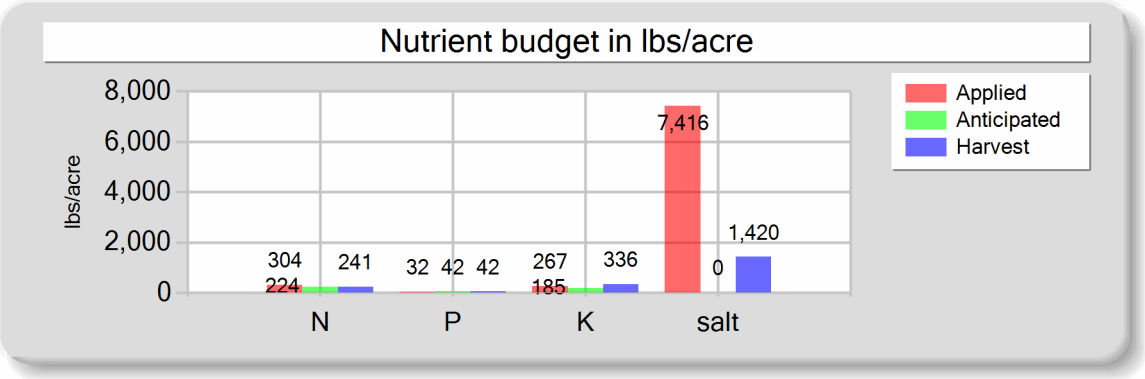
Fresh water applied	
12,960,000.00	*gallons*
477.27	*acre-inches*
9.55	*inches/acre*

Process wastewater applied	
3,540,500.00	*gallons*
130.38	*acre-inches*
2.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 06/16/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	223.60	31.59	266.68	1,251.33
Fresh water	38.21	0.00	0.00	6,164.62
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.01	31.59	266.68	7,415.95
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	241.11	42.30	335.58	1,420.43
Nutrient balance	62.90	-10.71	-68.90	5,995.52
Applied to removed ratio	1.26	0.75	0.79	5.22

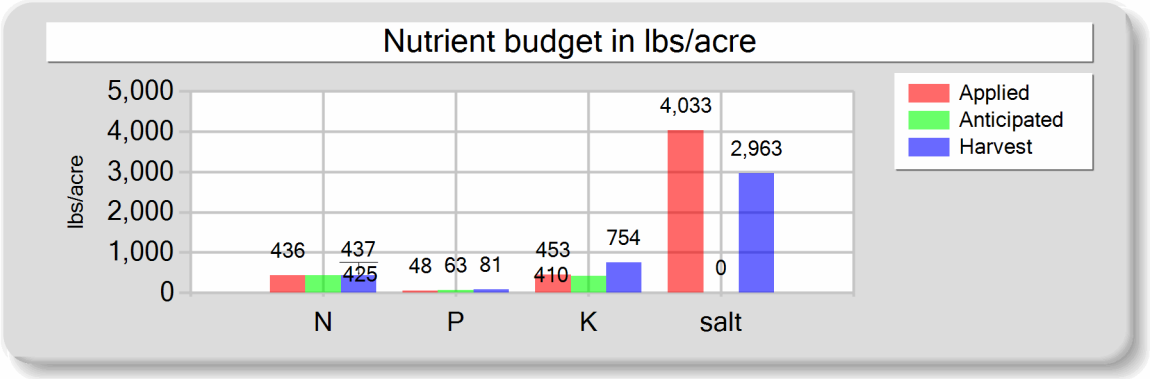
Fresh water applied	
38,475,000.00	*gallons*
1,416.90	*acre-inches*
28.34	*inches/acre*

Process wastewater applied	
6,145,500.00	*gallons*
226.32	*acre-inches*
4.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 10/08/2022: Rye Grass Silage

Field name: Field 7 Crop: Rye Grass Silage Plant date: 10/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	38.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	379.71	47.64	452.85	2,386.10
Fresh water	10.74	0.00	0.00	1,646.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	435.86	47.64	452.85	4,032.92
Anticipated crop nutrient removal	425.00	62.50	410.00	0.00
Actual crop nutrient removal	436.81	80.92	754.41	2,962.64
Nutrient balance	-0.95	-33.28	-301.56	1,070.28
Applied to removed ratio	1.00	0.59	0.60	1.36

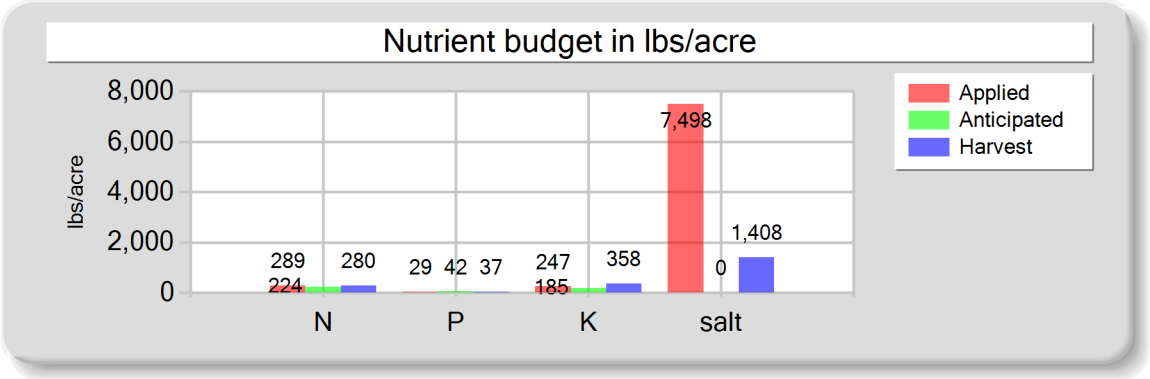
Fresh water applied	
19,725,000.00	*gallons*
726.40	*acre-inches*
9.95	*inches/acre*

Process wastewater applied	
4,661,780.00	*gallons*
171.68	*acre-inches*
2.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 06/17/2023: Corn, silage

Field name: Field 7 Crop: Corn, silage Plant date: 06/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	207.43	29.31	247.39	1,160.83
Fresh water	39.28	0.00	0.00	6,337.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	288.91	29.31	247.39	7,498.45
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	279.71	36.54	358.10	1,407.83
Nutrient balance	9.20	-7.23	-110.71	6,090.62
Applied to removed ratio	1.03	0.80	0.69	5.33

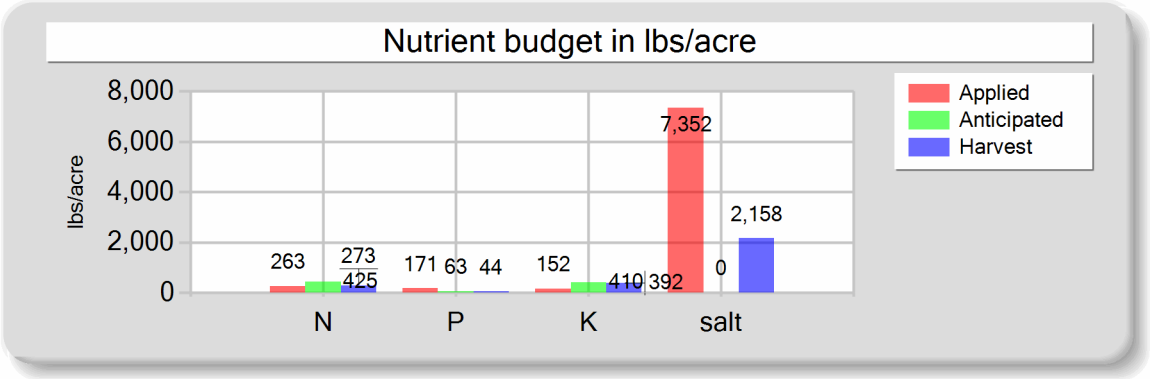
Fresh water applied	
57,750,000.00	*gallons*
2,126.74	*acre-inches*
29.13	*inches/acre*

Process wastewater applied	
8,323,460.00	*gallons*
306.52	*acre-inches*
4.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8 - 10/16/2022: Rye Grass Silage

Field name: Field 8 Crop: Rye Grass Silage Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	256.20	170.80	152.23	7,351.74
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	263.20	170.80	152.23	7,351.74
Anticipated crop nutrient removal	425.00	62.50	410.00	0.00
Actual crop nutrient removal	273.29	43.51	391.63	2,158.33
Nutrient balance	-10.10	127.28	-239.40	5,193.41
Applied to removed ratio	0.96	3.93	0.39	3.41

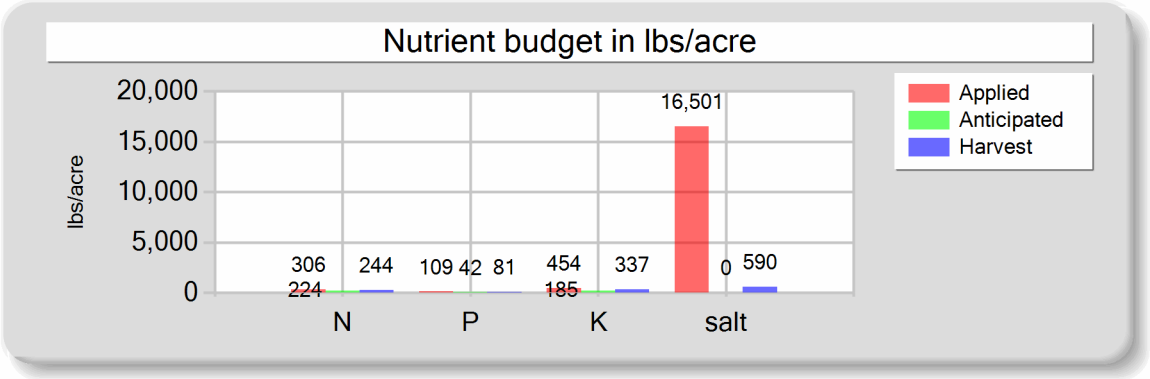
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 8 - 06/04/2023: Corn, silage

Field name: Field 8 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	35.20	0.00	0.00	0.00
Dry manure	247.74	109.06	453.74	8,239.97
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	15.67	0.00	0.00	8,261.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	305.61	109.06	453.74	16,501.22
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	244.48	80.65	337.19	590.30
Nutrient balance	61.14	28.41	116.55	15,910.92
Applied to removed ratio	1.25	1.35	1.35	27.95

Fresh water applied	
46,260,000.00	*gallons*
1,703.60	*acre-inches*
34.07	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Compost

Sample and source description: Compost

Sample date: 09/12/2022 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 37.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,400.00	7,460.00	42,200.00							51.40
DL	100.00	100.00	100.00							0.01

Compost

Sample and source description: Compost

Sample date: 04/17/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 21.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,400.00	4,000.00	12,400.00							54.20
DL	100.00	100.00	100.00							0.01

Compost

Sample and source description: Compost

Sample date: 04/17/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 21.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,400.00	4,000.00	12,400.00							54.20
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 38.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	8,100.00	33,700.00	11,900.00	9,100.00	4,900.00	5,600.00	2,210.00		61.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 38.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	8,100.00	33,700.00							61.20
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,700.00	13,800.00	12,300.00							59.40
DL	100.00	100.00	100.00							0.01

Compost

Sample and source description: Compost

Sample date: 10/09/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 32.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,900.00	10,700.00	14,700.00							55.90
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,900.00	10,000.00	46,000.00							44.70
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

1QWW

Sample and source description: 1QWW

Sample date: 02/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	885.00	378.00	0.00	0.00	138.00	878.00	207.00	160.00	160.00	2,780.00	2.00	406.00	361.00	6,180.00	4,300
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.50	1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	787.00	358.00	0.00	0.00	76.20	962.00								6,240.00	4,020
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	218.00	148.00	0.00	0.00	30.80	260.00								2,230.00	218
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	218.00	148.00	0.00	0.00	30.80	260.00								2,230.00	1,220
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/24/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	647.00	345.00	0.00	0.00	101.00	751.00								5,650.00	4,880
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	585.00	328.00	0.00	0.00	77.60	692.00								4,880.00	3,740
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Deep Well #10

Fresh Water

Sample description: Fresh Water

Sample date: 09/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.03	0.00	2.03								2,030.00	1,070
DL	0.50	0.50	0.50								1.00	10

Deep Well #15

Fresh Water

Sample description: Fresh Water

Sample date: 09/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.74	0.00	4.94								2,210.00	1,140
DL	0.50	0.50	0.50								1.00	10

District H20

District H20

Fresh Water

Sample description: Fresh Water

Sample date: 09/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.95	0.00	0.00								2,040.00	960
DL	0.50	0.50	0.50								1.00	10

IW 10

Fresh Water

Sample description: Fresh Water

Sample date: 10/24/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.39	0.00	0.00								234.00	190
DL	0.50	0.50	0.50								1.00	10

IW 13

Fresh Water

Sample description: Fresh Water

Sample date: 10/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.05	0.00	0.00								246.00	205
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Dondero 25-14 - 10/08/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass
Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 56.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	1,870.00	18,700.00		14.70
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass
Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 53.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,850.00	1,540.00	14,100.00		11.10
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass
Sample date: 06/26/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 77.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,860.00	670.00	7,400.00		14.50
DL	100.00	100.00	100.00		0.01

Dondero 25-14 - 06/27/2023: Corn, silage

Dondero 25-14 - 06/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 83.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,100.00	310.00	2,540.00		7.00
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,270.00	570.00	5,040.00		8.50
DL	100.00	100.00	100.00		0.01

Dondero 25-15 - 10/08/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass
Sample date: 05/01/2022 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,020.00	1,360.00	12,600.00		14.50
DL	100.00	100.00	100.00		0.01

Dondero 25-15 - 10/08/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,640.00	1,070.00	11,100.00		13.50
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass

Sample date: 06/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,990.00	980.00	9,820.00		18.90
DL	100.00	100.00	100.00		0.01

Dondero 25-15 - 07/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,760.00	690.00	3,890.00		6.20
DL	100.00	100.00	100.00		0.01

Dondero 25-15 - 07/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,510.00	570.00	3,640.00		6.50
DL	100.00	100.00	100.00		0.01

Dondero 25-19 - 10/22/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,760.00	600.00	6,120.00		11.10
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,280.00	600.00	5,620.00		10.90
DL	100.00	100.00	100.00		0.01

Dondero 25-19 - 06/09/2023: Corn, silage

Dondero 25-19 - 06/09/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,010.00	540.00	3,720.00		7.30
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,770.00	2,700.00	4,520.00		4.70
DL	100.00	100.00	100.00		0.01

Dondero 25-4 - 10/20/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,540.00	630.00	6,620.00		11.10
DL	100.00	100.00	100.00		0.01

Dondero 25-4 - 10/20/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,250.00	700.00	6,450.00		11.30
DL	100.00	100.00	100.00		0.01

Dondero 25-4 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,840.00	1,320.00	4,190.00		2.70
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,950.00	1,360.00	5,440.00		2.70
DL	100.00	100.00	100.00		0.01

Field 1 - 10/08/2022: Rye Grass Silage

Field 1 - 10/08/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,220.00	720.00	7,340.00		10.60
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,990.00	720.00	7,510.00		9.90
DL	100.00	100.00	100.00		0.01

Field 1 - 05/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,630.00	680.00	6,280.00		3.00
DL	100.00	100.00	100.00		0.01

Field 1 - 05/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,170.00	670.00	6,270.00		2.50
DL	100.00	100.00	100.00		0.01

Field 1S - 10/17/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,050.00	1,100.00	10,200.00		8.97
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,920.00	1,000.00	9,020.00		8.20
DL	100.00	100.00	100.00		0.01

Field 1S - 06/03/2023: Corn, silage

Field 1S - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,940.00	5,500.00	5,920.00		3.70
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,630.00	1,480.00	7,010.00		2.60
DL	100.00	100.00	100.00		0.01

Field 2 - 10/10/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass
Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,990.00	1,050.00	8,740.00		12.60
DL	100.00	100.00	100.00		0.01

Field 2 - 05/28/2023: Corn, silage

Field 2 - 05/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,970.00	680.00	6,480.00		2.10
DL	100.00	100.00	100.00		0.01

Field 3 - 10/08/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,380.00	850.00	8,790.00		10.50
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,800.00	1,830.00	16,800.00		13.30
DL	100.00	100.00	100.00		0.01

Field 3 - 10/08/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass

Sample date: 06/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,270.00	850.00	14,200.00		15.80
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass

Sample date: 06/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,110.00	1,360.00	18,400.00		9.40
DL	100.00	100.00	100.00		0.01

Field 3 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,550.00	830.00	3,730.00		3.40
DL	100.00	100.00	100.00		0.01

Field 3 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,620.00	620.00	4,620.00		5.40
DL	100.00	100.00	100.00		0.01

Field 3S - 10/18/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,970.00	700.00	5,480.00		9.48
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,540.00	1,180.00	9,700.00		10.20
DL	100.00	100.00	100.00		0.01

Field 3S - 06/01/2023: Corn, silage

Field 3S - 06/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,300.00	610.00	4,450.00		1.80
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,620.00	610.00	4,530.00		3.30
DL	100.00	100.00	100.00		0.01

Field 4 - 10/12/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass
Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,890.00	780.00	7,590.00		11.40
DL	100.00	100.00	100.00		0.01

Field 4 - 10/12/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,040.00	750.00	7,400.00		12.20
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass

Sample date: 06/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,780.00	1,470.00	20,600.00		12.60
DL	100.00	100.00	100.00		0.01

Field 4 - 06/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 80.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,340.00	320.00	2,440.00		5.30
DL	100.00	100.00	100.00		0.01

Field 4 - 06/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,670.00	640.00	4,510.00		5.20
DL	100.00	100.00	100.00		0.01

Field 5 - 10/15/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,170.00	950.00	8,820.00		11.90
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,990.00	1,170.00	10,700.00		11.20
DL	100.00	100.00	100.00		0.01

Field 5 - 10/15/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass
Sample date: 06/26/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 79.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,650.00	910.00	12,400.00		13.90
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass
Sample date: 06/26/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,330.00	1,140.00	13,300.00		12.90
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass
Sample date: 06/26/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 69.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,060.00	1,080.00	13,200.00		14.20
DL	100.00	100.00	100.00		0.01

Field 5 - 06/10/2023: Corn, silage

Field 5 - 06/10/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 59.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,070.00	1,200.00	6,780.00		1.30
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage
Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,630.00	890.00	6,070.00		2.50
DL	100.00	100.00	100.00		0.01

Field 6 - 10/08/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass
Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is
Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,950.00	760.00	8,100.00		10.60
DL	100.00	100.00	100.00		0.01

Field 6 - 10/08/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,350.00	790.00	7,780.00		11.20
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass

Sample date: 06/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,080.00	1,070.00	12,200.00		14.00
DL	100.00	100.00	100.00		0.01

Field 6 - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 53.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,590.00	850.00	8,070.00		2.70
DL	100.00	100.00	100.00		0.01

Field 6 - 06/16/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,420.00	600.00	4,760.00		6.90
DL	100.00	100.00	100.00		0.01

Field 7 - 10/08/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,470.00	1,660.00	11,200.00		11.50
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,020.00	930.00	8,670.00		12.80
DL	100.00	100.00	100.00		0.01

Field 7 - 10/08/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass

Sample date: 06/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,260.00	1,090.00	12,700.00		15.20
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass

Sample date: 06/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,110.00	980.00	10,300.00		12.00
DL	100.00	100.00	100.00		0.01

Field 7 - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,210.00	550.00	5,390.00		6.50
DL	100.00	100.00	100.00		0.01

Field 7 - 06/17/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/16/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,420.00	390.00	4,600.00		5.70
DL	100.00	100.00	100.00		0.01

Field 8 - 10/16/2022: Rye Grass Silage

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,150.00	820.00	7,380.00		12.40
DL	100.00	100.00	100.00		0.01

Rye Grass

Sample and source description: Rye Grass

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 74.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,060.00	700.00	5,610.00		13.70
DL	100.00	100.00	100.00		0.01

Field 8 - 06/04/2023: Corn, silage

Field 8 - 06/04/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,410.00	1,170.00	4,490.00		3.10
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,850.00	1,270.00	5,310.00		2.80
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

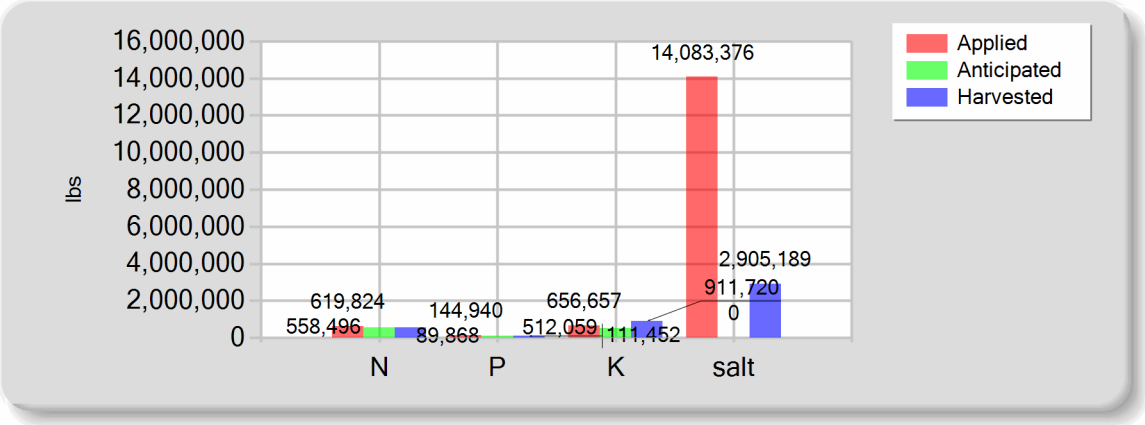
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

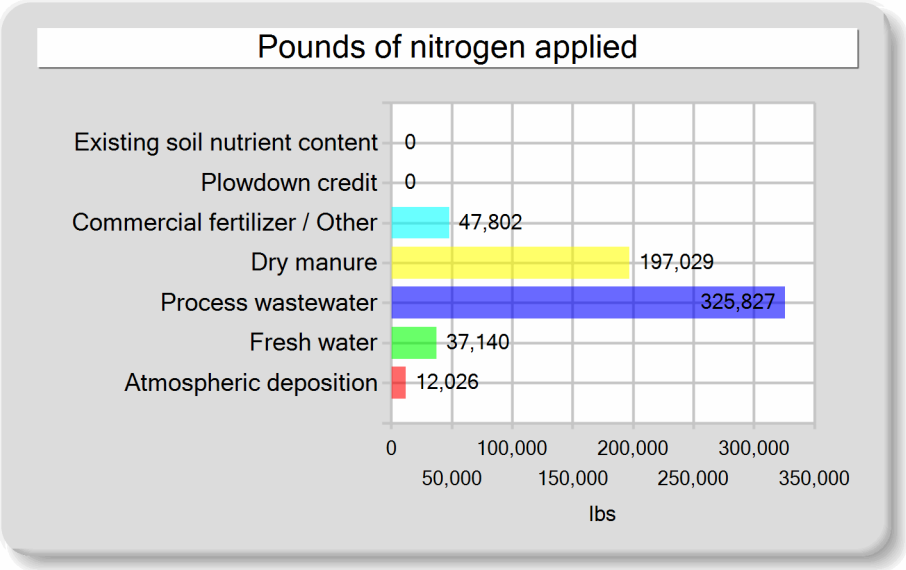
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

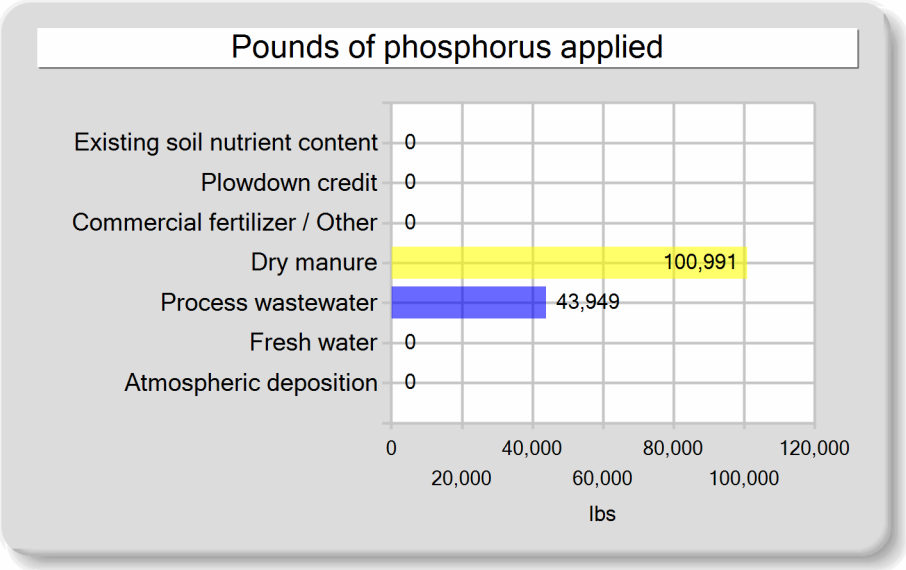
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	47,801.60	0.00	0.00	0.00
Dry manure	197,029.34	100,991.10	284,024.03	6,055,888.44
Process wastewater	325,826.78	43,949.09	372,633.18	1,811,854.56
Fresh water	37,140.20	0.00	0.00	6,215,633.22
Atmospheric deposition	12,026.00	0.00	0.00	0.00
Total nutrients applied	619,823.91	144,940.19	656,657.21	14,083,376.22
Anticipated crop nutrient removal	558,496.00	89,868.00	512,059.20	0.00
Actual crop nutrient removal	562,302.31	111,452.06	911,719.51	2,905,189.00
Nutrient balance	57,521.61	33,488.13	-255,062.30	11,178,187.22
Applied to removed ratio	1.10	1.30	0.72	4.85

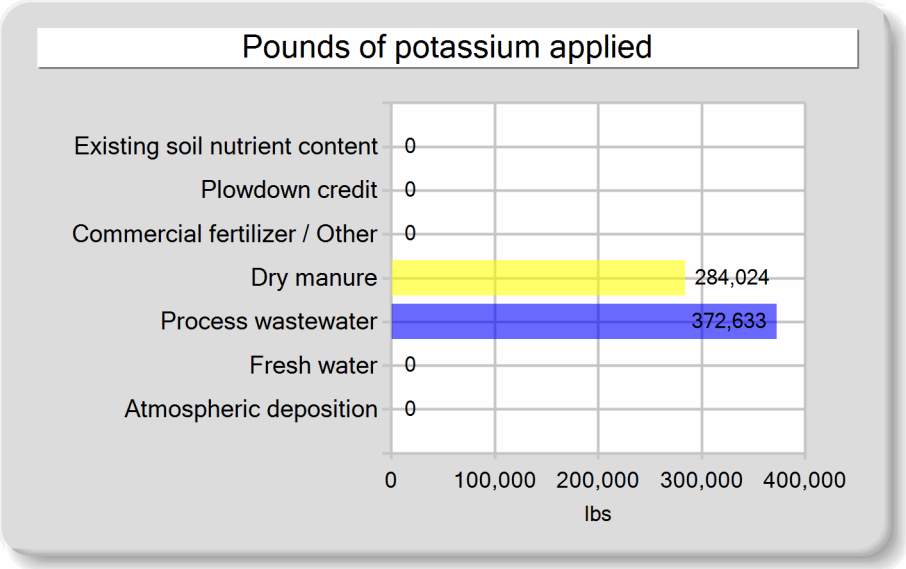
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

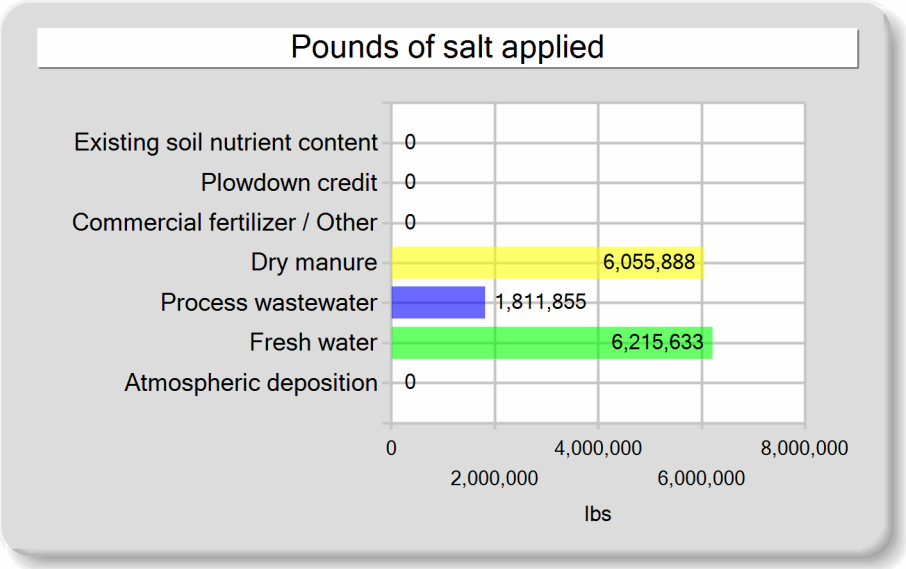


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field Dondero 25-14 Corn (1.50) had a high nitrogen field ratio as field failed to meet anticipated nitrogen removal rate of 224 pounds/acre. The subject field removed only 197 pounds of nitrogen/acre. Yield for the subject field was adequate however a low nitrogen value from the plant tissue lab analysis resulted in only 6.54 pounds of nitrogen removed per harvested ton compared to the anticipated 8 pounds of nitrogen removed per harvested ton which was ultimately the main reason for the high nitrogen field ratio. Nitrogen application to the field was not considered excessive but in-line with that recommended in the dairy's nutrient management plan.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Bert Weststeyn	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6/5/24	
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:
Name of Operator: Bert Weststeyn

Name of Dairy Facility: BWC Weststeyn Dairy, LP

Facility Address: 1763 S. Hewitt Road, Linden, CA 95236
Number and Street City Zip Code

Contact Person Name and Phone Number: Bert Weststeyn 209-404-9100
Name Phone Number

Manure/Process Wastewater Hauler Information:
Name of Hauling Company/Person: BWC Weststeyn Dairy, LP

Address of Hauling Company /Person: 1763 S. Hewitt Road, Linden, CA 95236
Number and Street City Zip Code

Contact Person: Bert Weststeyn 209-404-9100
Name Phone Number

Destination Information:
Composting Facility / Broker / Farmer / Other (identify) Fertilizer Company (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):
The Scott's Company 23390 E. Flood Street, Linden, CA 95236 209-887-3845
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:
23390 E. Flood Street, Linden, CA 95236
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 1/15/23 to 11/10/23

Amount Hauled:
Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 12,501 Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): 32.9% moisture
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

463 Loads at 27 ton per load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Bill Westlyf Date: 6/5/24

Hauler's Signature: Bill Westlyf Date: 6/5/24

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Authorized Copy Released To: _____

Client: BWC Weststein Dairy LP
Mailing address: 1763 S. Hewitt Road
(not facility address) Kingsburg, CA 95236
Phone#: 209-886-5334

Sampled By: _____
Project ID: _____

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- **H1** EC*, NO3-N, NH4-N*
- **H6** EC, TN, TDS
- **H7** Combination of H1 & H6
- **H14** EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- **H3** EC*, NH4-N, TKN, TP, K, TDS
- **H4** EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- **H5** Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- **H8** Combination of H3 &H5
- **H9** Combination of H4 &H5

TILE DRAIN
- **H2** EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- **E1** EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- **E2** EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- **E3** EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- **E4** EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- **M1** % Moisture
- **M3** TN TP, K, % Moisture
- **M4** Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- **M7** Combination of M3 & M4

PLANT TISSUE
- **F18** % Moisture, Ash, TN, P, K, DM and as received
- **F18+** % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- **F19+** % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- **K3** Soluble Phosphorus
- **K4** 1ft NO3, Organic Matter
- **K5** Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- **K6** 0-1ft EC, NO3, Soluble P, K, Organic Matter
- **K7** Combination of K3 & K4
- **K8** Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	9/13	AM	District H2O		17.6°						H6
2	"	"	Deep Well #15		19.8°						H7
3	"	"	Deep well # 10		18.8°						H7
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	(LABORATORY USE ONLY) LABORATORY NUMBER
1st	Rachel Weststein	BWC		9/13 1150 AM		
2nd	Albert Mortes Jr.	DENELE ANALYTICAL, INC.	9-13-23 11:51 AM			T3256016

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
BWC Weststeyn Dairy LP
1763 So. Hewitt Rd.
Linden, CA 95236

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/13/23 11:51 AM
SUBMITTED BY: Courier
ANALYZED DATE: 9/14/23

Irrigation/Ground Water (H6)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 17.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
District H20	T3256016A	Total Dissolved Solids (TDS)	SM 2540 C-1997 and	960	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	5.95	mg/L
		Electrical Conductivity (EC)	EPA 120.1 and	2.04	mmhos/cm

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Electrical Conductivity (EC)	95.0			2.04 (T3256016A)	2.07 (null)	1.36
Nitrate Nitrogen (NO3-N)	97.6	96.3 (T3255009B)	95.3 (T3255009B)			1.02
Total Dissolved Solids (TDS)	101			10.0 (T3262030A)	10.0 (Dup)	0.000
Total Kjeldahl Nitrogen (TKN)	98.6			2.80 (T3262030A)	2.80 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
BWC Weststeyn Dairy LP
1763 So. Hewitt Rd.
Linden, CA 95236

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/13/23 11:51 AM
SUBMITTED BY: Courier
ANALYZED DATE: 9/14/23

Irrigation / Ground Water (H7)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 19.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Deep Well #15	T3256016B	Electrical Conductivity (EC)	EPA 120.1 and	2.21	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.94	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	1140	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	7.74	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1410	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	98.0	98.2 (T3255009B)	96.4 (T3255009B)			1.85
Electrical Conductivity (EC)	95.0			2.04 (T3256016A)	2.07 (null)	1.36
Nitrate Nitrogen (NO3-N)	97.6	96.3 (T3255009B)	95.3 (T3255009B)			1.02
Total Dissolved Solids (TDS)	101			10.0 (T3262030A)	10.0 (Dup)	0.000
Total Nitrogen (TN)	98.6			2.80 (T3262030A)	2.80 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
BWC Weststeyn Dairy LP
1763 So. Hewitt Rd.
Linden, CA 95236

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/13/23 11:51 AM
SUBMITTED BY: Courier
ANALYZED DATE: 9/14/23

Irrigation / Ground Water (H7)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Deep Well #10	T3256016C	Electrical Conductivity (EC)	EPA 120.1 and	2.03	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.03	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	1070	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	2.03	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1300	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	98.0	98.2 (T3255009B)	96.4 (T3255009B)			1.85
Electrical Conductivity (EC)	95.0			2.04 (T3256016A)	2.07 (null)	1.36
Nitrate Nitrogen (NO3-N)	97.6	96.3 (T3255009B)	95.3 (T3255009B)			1.02
Total Dissolved Solids (TDS)	101			10.0 (T3262030A)	10.0 (Dup)	0.000
Total Nitrogen (TN)	98.6			2.80 (T3262030A)	2.80 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.
1232 SOUTH AVENUE
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: BWC Weststeyn Dairy LP
Mailing address: 1763 S. Hewitt Rd
(not facility address) Linden, CA 95236
Phone#: 209-886-5334

Authorized Copy Released To:

Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- **H1** EC*, NO3-N, NH4-N*
- **H6** EC, TN, TDS
- **H7** Combination of H1 & H6
- **H14** EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- **H3** EC*, NH4-N, TKN, TP, K, TDS
- **H4** EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- **H5** Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- **H8** Combination of H3 &H5
- **H9** Combination of H4 &H5

TILE DRAIN
- **H2** EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- **E1** EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- **E2** EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- **E3** EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- **E4** EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- **M1** % Moisture
- **M3** TN TP, K, % Moisture
- **M4** Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- **M7** Combination of M3 & M4

PLANT TISSUE
- **F18** % Moisture, Ash, TN, P, K, DM and as received
- **F18+** % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- **F19+** % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- **K3** Soluble Phosphorus
- **K4** 1ft NO3, Organic Matter
- **K5** Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- **K6** 0-1ft EC, NO3, Soluble P, K, Organic Matter
- **K7** Combination of K3 & K4
- **K8** Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	9/15/23	9:30 am	DW 30 HP	Domestic	21.8'		X				
2	9/15/23	9:15 am	DW 50 HP	Domestic	20.8'		X				
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	*(LABORATORY USE ONLY)* Logged in By:	LABORATORY NUMBER
1st	Cheryl Weststeyn	BWC Weststeyn				
2nd	[illegible]	DENELE ANALYTICAL, INC.	9-15-23 1:30			T3258045

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
BWC Weststeyn Dairy LP
1763 So. Hewitt Rd.
Linden, CA 95236

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/15/23 1:30 PM
SUBMITTED BY: Cheryl
ANALYZED DATE: 9/18/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 30 HP	T3258045A	Electrical Conductivity (EC)	EPA 120.1 and	0.261	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	3.56	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	167	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.2	94.0 (T3258022A)	94.6 (T3258022A)	0.635
Electrical Conductivity (EC)	97.6			1.87
Nitrate Nitrogen (NO3-N)	105	104 (T3258022A)	103 (T3258022A)	0.790

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
BWC Weststeyn Dairy LP
1763 So. Hewitt Rd.
Linden, CA 95236

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 9/15/23 1:30 PM
SUBMITTED BY: Cheryl
ANALYZED DATE: 9/18/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 20.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW 50 HP	T3258045B	Electrical Conductivity (EC)	EPA 120.1 and	0.251	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.28	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	161	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.2	94.0 (T3258022A)	94.6 (T3258022A)	0.635
Electrical Conductivity (EC)	97.6			1.87
Nitrate Nitrogen (NO3-N)	105	104 (T3258022A)	103 (T3258022A)	0.790

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714